

Highlights

Governing Greenwashing through Selling Formats and Blockchain Verification in Platform Supply Chains

- Reselling ordinary goods with green agency maximizes greenwashing at fees up to 50%
- More genuine greenness can be accompanied by more absolute exaggeration
- Verification can protect consumers while reducing total welfare
- Proportional exaggeration depends on relative costs, not selling formats
- Costless verification yields a common trust threshold for all members

Governing Greenwashing through Selling Formats and Blockchain Verification in Platform Supply Chains

Abstract

Greenwashing, in which suppliers exaggerate the environmental performance of their products, undermines consumer trust and distorts competition in platform supply chains. This paper studies how selling formats and blockchain verification shape greenwashing on e-commerce platforms. We develop a game-theoretic model in which an ordinary supplier and a green supplier sell through a platform, and the green supplier may exaggerate its greenness because consumers cannot verify environmental claims. We compare four selling-format configurations—both reselling (RR), ordinary reselling with green agency (RA), ordinary agency with green reselling (AR), and both agency (AA)—with and without blockchain, and then endogenize both genuine greenness and exaggeration. First, pricing authority and revenue sharing change the supplier’s profit from exaggeration: RA never produces the least greenwashing and produces the most when the commission rate does not exceed one half, so the platform’s most profitable format can differ from the format with the least greenwashing. Second, verification restores credibility but also removes demand created by exaggerated claims. It can protect consumers while lowering total welfare, or raise total welfare while lowering realized consumer surplus. Third, when both are chosen jointly without verification, proportional exaggeration depends only on the relative costs of green investment and exaggeration, whereas selling format and consumer trust determine genuine greenness and hence the absolute amount exaggerated. Higher genuine greenness can therefore come with more greenwashing. With costless verification, all supply chain members switch their adoption preference at the same trust threshold. Platforms should therefore evaluate green investment, greenwashing, and welfare separately.

Keywords: Greenwashing, Blockchain technology, Platform supply chain, Selling format, Game theory

1. Introduction

Consumers increasingly prefer products with better environmental performance (Ye et al. 2024). Survey data show that more than 80% of consumers are willing to pay a premium for sustainably produced goods, with an average stated premium of 9.7%¹. This demand gives firms strong incentives to market products as green, but it also rewards greenwashing: exaggerating a product’s environmental performance beyond what the firm has actually achieved.

When consumers cannot verify environmental claims, firms can capture green price premiums without making the corresponding environmental investment (Huang et al. 2020, Wang et al. 2025). Common practices include selective disclosure, vague environmental language, and unverifiable eco-labels (Chen and Duan 2023, Xia et al. 2026). Recent cases show the practical relevance of this problem. In the fashion industry, the UK’s Advertising Standards Authority banned advertisements from Nike and Superdry in December 2025 after ruling that their environmental claims were misleading—Nike had described polo shirts as using “sustainable materials” without qualification², while Superdry promoted “Sustainable Style” without supporting evidence³. In the automotive sector, Mercedes-Benz faced criticism for associating its electric vehicles with images of pristine nature⁴. These cases reflect a basic information problem: consumers often cannot observe a product’s true environmental performance (Dong et al. 2023a). Capgemini’s 2025 consumer data report that 63% of shoppers lack enough information to verify sustainability claims and 54% do not trust such claims⁵. Such distrust makes it harder for genuinely green products to distinguish themselves and weakens incentives for real environmental investment (Gao et al. 2025).

The problem is particularly relevant in online platform supply chains, where the platform decides how competing suppliers sell to consumers. The first decision is the selling format: under reselling, the platform buys the product at a wholesale price and sets the retail price;

¹PwC 2024 Voice of the Consumer Survey

²ASA ruling on Nike

³ASA ruling on Superdry

⁴<https://www.marketing-interactive.com/mercedes-benz-distances-greenwashing-ad>

⁵Capgemini consumer report, 2025

under agency, the supplier sets the retail price and pays a commission to the platform (Abhishek et al. 2016, Tian et al. 2018). In practice, JD.com uses both self-operated reselling and third-party agency selling (Yuan and Bao 2025, Zhao et al. 2024). These formats allocate pricing authority and revenue between the platform and each supplier (Wu et al. 2025, Wu and Zhao 2027). They therefore also change how much the green supplier earns from an exaggerated environmental claim. The platform chooses the format that maximizes its profit, which may differ from the format that produces the least greenwashing.

The second decision is whether to adopt blockchain-enabled traceability (Xu et al. 2025, Wu et al. 2025). Alibaba’s Tmall Global, for example, uses blockchain-based origin traceability for imported goods, allowing consumers to check information on origin, shipping, warehousing, and customs (Cao and Shen 2022). A similar “green organic + blockchain + traceability + standards” model has been used in China’s Sichuan Province⁶. We model blockchain as full verification backed by credible source data, so consumers observe true greenness and exaggerated claims are removed (Shen et al. 2022, Wu and Zhao 2024). Verification also creates a per-unit cost for the platform (Dong et al. 2023a, Wu et al. 2025). The platform adopts it only when the benefit from verified information is large enough to cover this cost and the loss of sales generated by exaggerated claims.

The closest benchmark is Wu et al. (2025), who study the same four selling-format configurations and blockchain transparency but assume truthful environmental claims. They find that RA promotes genuine greenness at low commissions, leaving open whether the same contract also curbs exaggeration and how removing inflated claims changes demand and welfare. Xu et al. (2025) examine greenwashing and blockchain with one upstream manufacturer; our setting adds competition between green and ordinary suppliers whose selling formats can differ. These differences let us identify when a greenness-promoting contract encourages greenwashing and when verification shifts consumer and welfare outcomes in opposite directions.

To address these gaps, this study seeks to answer the following three research questions:
 (1) In the absence of blockchain, how do different selling format configurations affect the

⁶<http://kjbs.sckjw.com.cn/00d9b4327099433aa2e0cb421631990e>

green supplier’s greenwashing level, and which configuration is most effective at curbing greenwashing?

(2) Under what conditions does the platform adopt blockchain technology to eliminate greenwashing, and does its choice protect consumers and raise total welfare?

(3) When the green supplier jointly chooses genuine greenness and the exaggeration level, how do selling formats and verification affect these two decisions, and when do supply chain members prefer blockchain adoption?

To answer these questions, we develop a game-theoretic model of a platform supply chain with two competing suppliers: one offering an ordinary product and the other a green product. The green supplier can engage in costly greenwashing to inflate consumers’ perceived greenness of his product, exploiting consumers’ inability to verify environmental claims. The platform chooses a selling format for each supplier—reselling or agency—giving rise to four possible configurations, and decides whether to adopt blockchain technology to provide verifiable proof of the green product’s true environmental performance. We first treat genuine greenness as given and then allow the green supplier to choose both genuine greenness and exaggeration. We characterize the equilibrium greenwashing level, prices, and profits, and identify when blockchain adoption benefits consumers, the platform, and, once greenness is endogenous, every supply chain member.

The paper makes three contributions. First, we show that selling formats alter the green supplier’s return to exaggeration through pricing authority and revenue sharing (Proposition 2). At low commission rates, RA produces the highest genuine greenness with verification, but without verification it produces the most greenwashing when the commission rate does not exceed one half and never the least (Propositions 2 and 5). A contract favored for greenness or platform profit can therefore worsen misleading claims. Second, when claims may be inflated, verification both corrects consumers’ beliefs and removes demand created by exaggeration (Propositions 3 and 4). Under market power, these effects can oppose each other, so platform adoption, realized consumer surplus, and total welfare can favor different choices. Third, when the green supplier chooses genuine greenness and exaggeration jointly, proportional exaggeration depends only on their relative costs (Proposition 6). Selling format and consumer trust determine genuine greenness and hence absolute exaggeration, allowing

greenness and greenwashing to rise together. With costless verification, all supply chain members switch their adoption preference at the same trust threshold.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces the model. Section 4 presents the equilibrium analysis and optimal strategies. Section 5 endogenizes greenness and characterizes the supplier’s choice between genuine greenness and exaggeration. Section 6 concludes with managerial implications and directions for future research.

2. Literature Review

Our work connects three streams of literature: greenwashing in supply chains, platform selling format selection, and blockchain adoption in operations management. Together, they clarify how an intermediary’s contracts and verification decisions shape the supplier’s incentives to invest in genuine greenness and exaggerate its claims.

2.1. *Greenwashing in supply chains*

Consumers increasingly favor high-quality green products. However, information asymmetry has fueled corporate greenwashing, thereby distorting green product markets (Chen and Duan 2023, Ye et al. 2024). Existing research has largely focused on how to regulate and curb greenwashing, with some studies employing evolutionary game theory to explore this issue. For instance, Gao et al. (2025) analyze the dynamic evolution of firms’ greenwashing behavior under different regulatory environments within a tripartite game framework that includes new energy vehicle parts suppliers, manufacturers, and the government. Wang et al. (2025) construct an evolutionary game model involving firms, media, government, and consumers, clarifying the interest chain through which firms engage in bribery to induce these parties to participate in greenwashing. Hao et al. (2026) develop an evolutionary game model consisting of firms, investors, and rating agencies, concluding that mandatory ESG disclosure is a powerful measure to curb greenwashing. Nevertheless, the welfare implications of greenwashing are not uniformly negative. For instance, Lee et al. (2018) find that as long as a certain proportion of informed consumers exists in the market, allowing greenwashing may actually incentivize some firms to genuinely go green. Wu et al. (2020) show that

sufficiently high transparency can eliminate greenwashing, but when transparency increases further, socially responsible firms tend to reduce observable investments. [Huang et al. \(2020\)](#) discover that when the green gap in the market is small, greenwashing can benefit both incumbent green firms and consumer surplus. [Wei et al. \(2026\)](#) study competition between a socially responsible firm and a profit-maximizing firm that may greenwash through observable CSR activities or CSR advertising; under low transparency and intense competition, the responsible firm over-invests to deter greenwashing, so greenwashing can raise CSR activity and social welfare. These studies consider firms that sell directly to consumers and do not examine how an intermediary’s selling contracts change the return to misleading claims.

Several studies examine blockchain as a way to reduce greenwashing. [Dong et al. \(2023a\)](#) show that high adoption costs can prevent blockchain from eliminating greenwashing. [Xia et al. \(2026\)](#) study how blockchain disclosure of inventory transactions and carbon emissions affects greenwashing by capital-constrained firms. [Xu et al. \(2025\)](#) study blockchain adoption and endogenous greenwashing on live-streaming platforms. Their model has a single upstream manufacturer and does not jointly choose genuine greenness and exaggeration. Our model adds upstream competition and lets the green supplier choose both genuine greenness and exaggeration.

2.2. Platform selling format selection

A large body of literature has examined platforms’ choices between agency and reselling formats. Many studies focus on one-to-one or one-to-two supply chain structures ([Abhishek et al. 2016](#), [Ha et al. 2022](#), [Zha et al. 2023](#), [Hsiao et al. 2024](#)). We consider a two-to-one structure in which competing upstream suppliers can face different selling formats. For example, [Cheng et al. \(2023\)](#) examine the impact of a platform’s introduction of its own brand on existing suppliers under different selling formats. [Jiang et al. \(2024\)](#) examine whether upstream competing suppliers offer rebates under different selling formats. However, these two studies do not consider the hybrid format, in which the platform adopts different cooperative selling formats with the two suppliers. [Tian et al. \(2018\)](#) argue that the interaction between order fulfillment costs and the intensity of upstream competition influences a downstream platform’s optimal choice of selling formats. [Matsui \(2024\)](#) investigate whether two

competing suppliers choose to sell through direct channels, platform channels, or both, and whether the platform adopts an agency or reselling format. [Li et al. \(2024\)](#) integrate information sharing with selling format choices, finding that the platform always has an incentive to share information with direct-selling manufacturers; however, under a hybrid selling format, the platform may also share information with indirect-selling manufacturers. [Yuan and Bao \(2025\)](#) explore, in the presence of non-deceptive imitation products, how genuine and imitation suppliers should select their respective selling formats with the platform.

Most of the studies above do not endogenize product greenness. Closest to our structure, [Wu et al. \(2025\)](#) examine blockchain transparency, asymmetric selling contracts, competition between green and ordinary suppliers, and endogenous greenness. In their model, the green supplier’s attributes are reported truthfully and blockchain affects how consumers value genuine greenness. [Wu and Zhao \(2027\)](#) study how green R&D cost sharing and the platform’s private market information shape competing manufacturers’ investment and selling-mode choices, including a hybrid format; they do not model claim verification or greenwashing.

Our study adds exaggeration as a supplier decision to the truthful-attribute platform setting. The green supplier chooses how much to overstate an unverifiable environmental attribute, so consumers may overvalue or undervalue true greenness depending on trust. This changes three results. First, RA can promote genuine greenness when attributes are truthful ([Wu et al. 2025](#)), but at commissions up to one half it produces the most greenwashing in our model (Proposition 2). The same contract can therefore strengthen both genuine investment and exaggeration. Second, verification restores accurate information but removes demand generated by exaggerated claims, so the platform’s adoption decision, realized consumer surplus, and total welfare can favor different choices (Propositions 3 and 4). Third, when genuine greenness and exaggeration are jointly chosen, relative costs determine proportional exaggeration, while selling format changes genuine greenness and therefore the total amount exaggerated (Proposition 6). Relative to [Xu et al. \(2025\)](#), our model adds upstream competition and asymmetric selling formats to the joint analysis of genuine environmental investment and exaggeration.

2.3. Blockchain in operations management

In the existing literature on blockchain, some studies focus on its traceability to clarify the responsibilities of each participant (Iyengar et al. 2024, Cui et al. 2023, Dong et al. 2023b), while others emphasize its transparency to facilitate information sharing between upstream and downstream parties (Cui et al. 2024, Keskin et al. 2025). This study concentrates on the role of blockchain in revealing the true quality of products. For example, Pun et al. (2021) and Shen et al. (2022) apply blockchain to combat counterfeit products. Zhang et al. (2022) analyze whether two retailers should adopt blockchain and how this decision is affected by enhanced information transparency and consumer privacy concerns. Zhang et al. (2023a) explore the bidirectional interaction between a manufacturer’s blockchain adoption strategy and gray market entry decisions in a global supply chain. Zhang et al. (2023b) focus on the impact of blockchain adoption on the decisions and efficiency of supply chain members in a dual-channel supply chain, with particular attention to members’ risk-averse characteristics. Wu and Zhao (2024) examine how blockchain technology can help a new entrant with green products successfully enter the market and compete with an incumbent selling ordinary products.

Some studies integrate blockchain application with the choice of selling formats. For instance, Zhao et al. (2024) find that manufacturers are more motivated to join a blockchain system under an agency mode than under a reselling mode. Chen et al. (2025) investigate whether a platform should introduce blockchain to enhance the quality credibility of refurbished products when facing competition between new and refurbished products, and analyze the effects of blockchain introduction under different selling formats for new products. Furthermore, research has also focused on green supply chains. Xu et al. (2023) explore optimal decisions and supply chain coordination when a manufacturer sells on a platform under different selling formats, and find that blockchain can improve product greenness and promote supply chain coordination. Wu et al. (2025) focus on the role of platforms in the competition between green and non-green suppliers, analyzing how different selling formats and blockchain technology affect product greenness. Table 1 summarizes the comparison with our study.

Table 1 summarizes the closest literature. We study whether the platform’s preferred

Table 1: Comparison between this study and existing literature

	Selling format	Upstream competition	Blockchain adoption decision	Greenwashing
Xu et al. (2023)	Reselling/Agency	×	×	×
Dong et al. (2023a)	Reselling	×	✓	✓
Zhao et al. (2024)	Reselling/Agency	×	✓	×
Xu et al. (2025)	Reselling/Agency	×	✓	✓
Chen et al. (2025)	Reselling/Agency	✓	✓	×
Wu et al. (2025)	Hybrid	✓	✓	×
Wu and Zhao (2027)	Hybrid	✓	×	×
Yuan and Bao (2025)	Hybrid	✓	×	×
Xia et al. (2026)	Direct selling	×	×	✓
Wei et al. (2026)	Direct selling	✓	×	✓
This study	Hybrid	✓	✓	✓

selling format also reduces greenwashing and how verification changes this comparison in a market with competing suppliers.

3. Model Description

We consider a platform supply chain consisting of three players: an ordinary product supplier (denoted by subscript o ; she), a green product supplier (denoted by g ; he), and an e-commerce platform (denoted by p ; it). Let subscript $m \in \{o, g, p\}$ index the three players. The platform serves as an intermediary between the suppliers and end consumers, and can choose between two selling formats for each supplier: reselling (R) or agency selling (A). In the reselling format, the platform purchases the product from the supplier m at a wholesale price w_m and then sets the retail price p_m . In the agency format, the supplier directly sets the retail price p_m and pays the platform a fraction $\lambda \in (0, 1)$ of the revenue as a commission fee. We assume that the commission rate λ is exogenously determined and is symmetric across suppliers when the agency format is adopted (Li et al. 2024, Yuan and Bao 2025). The combination of selling formats for the two suppliers yields four possible configurations:

(i) RR: both suppliers operate under reselling; (ii) RA: ordinary supplier under reselling, green supplier under agency; (iii) AR: ordinary supplier under agency, green supplier under reselling; and (iv) AA: both suppliers under agency.

We consider a green supplier whose product has a true greenness level $g > 0$ (e.g., environmental performance, carbon reduction) and incurs a cost of $\frac{k}{2}g^2$, where $k > 0$ is the green investment cost coefficient (Shen et al. 2022, Xu et al. 2023). In the basic model, to isolate greenwashing behavior, we treat greenness as exogenously given, for instance determined by technology standards or external certification (Xu et al. 2025). In Section 5, we relax this assumption and allow the green supplier to choose both the greenness level g and the extent of greenwashing.

The greenwashing level $\beta \geq 0$ captures the degree to which the supplier exaggerates its green claims. Through greenwashing, the declared greenness is amplified from g to $(\beta + 1)g$. Because consumers cannot verify these claims in the no-blockchain benchmark, their trust is discounted by a factor $\gamma \in (0, 1)$. Consequently, the effective perceived greenness by consumers is $\gamma(\beta + 1)g$. Unlike genuine investment, a higher β changes claims without improving true greenness. Greenwashing is costly for the supplier, encompassing potential costs from consumer returns when products are found to be less green than claimed, reputational damage, and penalties from regulators or other stakeholders (Wu et al. 2020, Huang et al. 2020). We capture these costs through a quadratic cost function $\theta(\beta g)^2$, where $\theta > 0$ is the greenwashing cost coefficient. These costs are borne by the green supplier.

Under blockchain adoption, on-chain traceability is combined with credible verification of source data, and consumers accept the verified information. Claim inflation is therefore eliminated ($\beta = 0$) and the trust discount disappears ($\gamma = 1$), so perceived greenness equals true greenness g (Shen et al. 2022, Wu and Zhao 2024). The platform incurs a per-unit cost $c_g \geq 0$ for this system (e.g., technology infrastructure and verification fees) (Wu et al. 2025).

Consumers are heterogeneous in their environmental preferences. The market consists of a mass $a > 0$ of potential consumers, of whom a fraction $\rho \in (0, 1)$ are “ordinary-oriented” (preferring the ordinary product) and the remaining fraction $1 - \rho$ are “green-oriented” (preferring the green product). We assume a cross-price coefficient $b \in (0, 1)$ so that own-price effects dominate substitution effects. The demand functions are linear in prices and

perceived greenness. Table 2 summarizes all the symbols used in this study.

Table 2: Summary of notation

Parameters	Definition
Subscript m	o : ordinary supplier, g : green supplier, p : platform
Superscript i	N : no blockchain, B : blockchain adoption
Superscript j	RR, RA, AR, AA : selling format configurations
a	Fundamental market demand
ρ	Ordinary product market scale
b	Cross-price sensitivity between products
γ	Consumers' trust in green claims without blockchain
g	Greenness of the green product
k	Cost coefficient of green investment
β	Proportional inflation of true greenness in the supplier's claim
θ	Cost coefficient of greenwashing
λ	Commission rate (platform's revenue share in agency format)
c_g	Per-unit cost of blockchain adoption
p_m	Retail price of product m
w_m	Wholesale price of product m (reselling mode)
D_m	Demand for product m

3.1. No-blockchain scenario

In the absence of blockchain ($i = N$), consumers cannot verify the true greenness of the green product. They form their purchase decisions based on the perceived greenness, which is inflated by greenwashing: $\gamma(\beta + 1)g$. The demand functions are:

$$D_g^N = (1 - \rho)a - p_g + bp_o + \gamma(\beta + 1)g, \quad (1)$$

$$D_o^N = \rho a - p_o + bp_g. \quad (2)$$

Greenness shifts the green product's demand intercept, while the ordinary product is affected indirectly through price competition. This specification captures green demand expansion rather than pure cannibalization of the ordinary market.

At the outset, the platform chooses the selling format for each supplier and whether to adopt blockchain by comparing its equilibrium profits across the feasible scenarios. For a given selling format configuration, we model the subsequent decision sequence as a two-stage game:

- (1) Stage 1 (Supplier decisions): The two suppliers move simultaneously. A supplier under reselling sets the wholesale price w_m ; a supplier under agency sets the retail price p_m directly. At the same time, the green supplier chooses the greenwashing level β .
- (2) Stage 2 (Platform pricing): Observing the suppliers' decisions, the platform sets the retail price p_m of each product sold under reselling.

The timing treats greenwashing as a contemporaneous marketing decision rather than a prior commitment. Agency sellers post retail prices before the platform prices resold products, reducing to the standard RR and AA games in the symmetric cases. We compare only feasible equilibria with positive decisions, satisfied second-order conditions, and nonnegative member profits. The profit functions for the three players in each selling format are as follows.

Case NRR (both reselling):

$$\pi_o^{NRR} = w_o D_o^N, \quad (3)$$

$$\pi_g^{NRR} = w_g D_g^N - \frac{k}{2} g^2 - \theta(\beta g)^2, \quad (4)$$

$$\pi_p^{NRR} = (p_g - w_g) D_g^N + (p_o - w_o) D_o^N. \quad (5)$$

Case NRA (ordinary reselling, green agency):

$$\pi_o^{NRA} = w_o D_o^N, \quad (6)$$

$$\pi_g^{NRA} = (1 - \lambda) p_g D_g^N - \frac{k}{2} g^2 - \theta(\beta g)^2, \quad (7)$$

$$\pi_p^{NRA} = \lambda p_g D_g^N + (p_o - w_o) D_o^N. \quad (8)$$

Case NAR (ordinary agency, green reselling):

$$\pi_o^{NAR} = (1 - \lambda) p_o D_o^N, \quad (9)$$

$$\pi_g^{NAR} = w_g D_g^N - \frac{k}{2} g^2 - \theta(\beta g)^2, \quad (10)$$

$$\pi_p^{NAR} = (p_g - w_g)D_g^N + \lambda p_o D_o^N. \quad (11)$$

Case NAA (both agency):

$$\pi_o^{NAA} = (1 - \lambda)p_o D_o^N, \quad (12)$$

$$\pi_g^{NAA} = (1 - \lambda)p_g D_g^N - \frac{k}{2}g^2 - \theta(\beta g)^2, \quad (13)$$

$$\pi_p^{NAA} = \lambda p_g D_g^N + \lambda p_o D_o^N. \quad (14)$$

3.2. Blockchain adoption scenario

Under blockchain adoption ($i = B$), verified true greenness replaces unverified claims. Consequently, $\beta = 0$, and the greenwashing cost term $\theta(\beta g)^2$ vanishes. However, the platform incurs a per-unit blockchain cost c_g for the green product. The demand functions remain unchanged for the ordinary product, while the green product demand becomes:

$$D_g^B = (1 - \rho)a - p_g + bp_o + g, \quad (15)$$

where the coefficient of g is now 1 rather than $\gamma(\beta + 1)$ because consumers fully trust the verified greenness and greenwashing is impossible. The profit functions of the green supplier and platform reflect the removal of β and the addition of c_g :

Case BRR:

$$\pi_g^{BRR} = w_g D_g^B - \frac{k}{2}g^2, \quad (16)$$

$$\pi_p^{BRR} = (p_g - w_g - c_g)D_g^B + (p_o - w_o)D_o^B. \quad (17)$$

Case BRA:

$$\pi_g^{BRA} = (1 - \lambda)p_g D_g^B - \frac{k}{2}g^2, \quad (18)$$

$$\pi_p^{BRA} = (\lambda p_g - c_g)D_g^B + (p_o - w_o)D_o^B. \quad (19)$$

Case BAR:

$$\pi_g^{BAR} = w_g D_g^B - \frac{k}{2}g^2, \quad (20)$$

$$\pi_p^{BAR} = (p_g - w_g - c_g)D_g^B + \lambda p_o D_o^B. \quad (21)$$

Case BAA:

$$\pi_g^{BAA} = (1 - \lambda)p_g D_g^B - \frac{k}{2}g^2, \quad (22)$$

$$\pi_p^{BAA} = (\lambda p_g - c_g)D_g^B + \lambda p_o D_o^B. \quad (23)$$

Blockchain changes the ordinary supplier's equilibrium payoff through prices and demand, but it does not change the form of that supplier's profit function. It retains the wholesale-revenue or agency-revenue structure in Equations (3), (6), (9), and (12), with D_o^B replacing D_o^N .

4. Equilibrium Analysis

We solve the game by backward induction. The full analytical expressions of prices, greenwashing levels, and profits are given in Appendix A, and proofs of all lemmas and propositions are provided in Appendix B.

4.1. No-blockchain scenario

4.1.1. Equilibrium properties

In the absence of blockchain technology, consumers cannot verify the true greenness of the green product, creating an opportunity for the green supplier to engage in greenwashing. The following lemma establishes the parametric conditions required to guarantee positive decision variables.

Lemma 1. *An interior equilibrium with positive prices and $\beta^{Nj} > 0$ exists, and satisfies the green supplier's second-order conditions, if and only if $\theta > \bar{\theta}^{Nj}$, where $\bar{\theta}^{NRR} = \frac{\gamma^2}{2(4-b^2)}$, $\bar{\theta}^{NRA} = \frac{2\gamma^2(1-\lambda)}{8-b^2(3\lambda+5)}$, $\bar{\theta}^{NAR} = \frac{\gamma^2(4-b^2(3\lambda+1))}{4(8-b^2(3\lambda+5))}$, and $\bar{\theta}^{NAA} = \frac{\gamma^2(1-\lambda)}{4-b^2}$.*

We restrict attention to parameter values satisfying $\theta > \bar{\theta}^{Nj}$ for each configuration being compared. The threshold is independent of g , rises with γ and b , and falls with λ in agency configurations.

Lemma 2. *In every configuration, proportional exaggeration β^{Nj} decreases in g , ρ , and θ and increases in γ and b . Absolute exaggeration $\beta^{Nj}g$ increases in g . For $j \in \{RA, AR, AA\}$, β^{Nj} also decreases in λ ; β^{NRR} does not depend on λ .*

Higher g lowers proportional exaggeration, but raises absolute exaggeration; higher γ increases the demand return to exaggeration, while stronger substitutability b raises differentiation incentives. A larger ordinary market share and a higher greenwashing cost reduce

this return. In configurations involving agency, a higher commission also weakens the incentive to exaggerate.

Proposition 1. *In every configuration, all wholesale prices, retail prices, and members' profits increase in γ .*

Higher trust increases both greenwashing and all members' profits. Greater trust makes the green product more attractive to consumers, which supports higher prices and demand. When greenwashing is inexpensive, the supplier can also obtain a price premium that is not supported by genuine greenness alone.

4.1.2. Impact of selling formats on greenwashing

Proposition 2. *Suppose all four no-blockchain configurations have interior equilibria, and let $x = a(1 - \rho) + \gamma g$ and $y = a\rho$. Define*

$$\hat{\lambda} = \frac{4b(2 + b^2)x + b^2(12 - \gamma^2/\theta)y}{12b(2 - b^2)x + [4(8 - 5b^2) - (4 - 3b^2)\gamma^2/\theta]y},$$

which satisfies $\hat{\lambda} < 1/2$ if $b \leq \sqrt{2/5}$.

(1) RA never minimizes greenwashing. For $\lambda \leq 1/2$, the minimum is attained by RR if $\lambda < \hat{\lambda}$ and by AR if $\lambda > \hat{\lambda}$; for $\lambda \geq 1/2$, it is attained by AA or AR.

(2) For $\lambda \leq 1/2$, RA maximizes greenwashing.

The ranking follows from how each format changes the green supplier's return to exaggeration. The $1/2$ boundary has a direct revenue-sharing interpretation. Under RR, the green supplier retains the full wholesale margin but faces double marginalization; under AA, he controls the retail price but keeps only $1 - \lambda$ of sales revenue. Their greenwashing incentives therefore reverse at $\lambda = 1/2$. The threshold $\hat{\lambda}$ compares RR with AR. Because the green product is resold in both formats, greenwashing moves with the green wholesale price. In AR, a higher commission on the ordinary product gives the platform a stronger incentive to steer demand toward that product, which lowers the green supplier's wholesale return to exaggeration. Hence AR overtakes RR as the least-greenwashing format once $\lambda > \hat{\lambda}$.

RA works in the opposite direction at low commissions. The green supplier combines direct pricing control with a large retained revenue share, while the rival product still carries

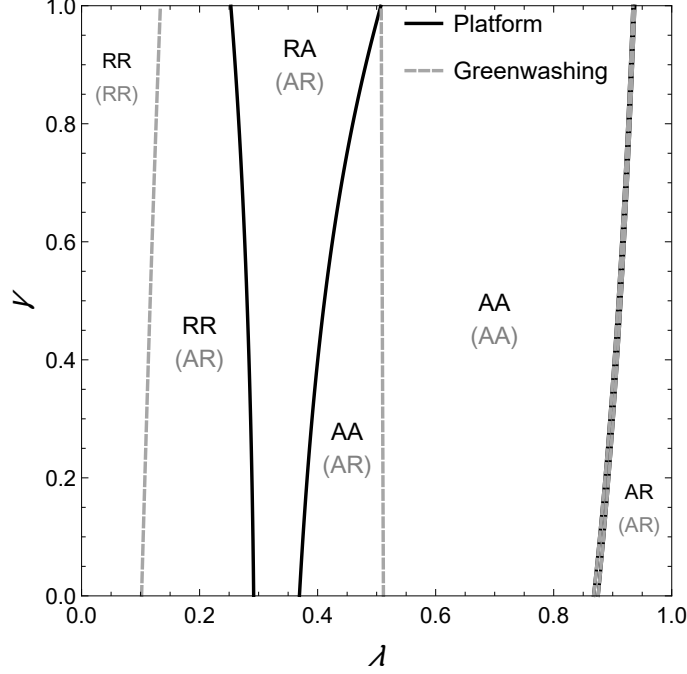


Figure 1: Selling format and greenwashing ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 0.4$, $g = 0.3$).

the platform's resale markup; this softens competition and raises the payoff from inflated claims. This explains why RA never minimizes greenwashing and, for $\lambda \leq 1/2$, maximizes it. As λ rises, the retained agency share shrinks and this incentive weakens. Relative to the one-manufacturer setting of [Xu et al. \(2025\)](#), upstream competition and asymmetric format assignment are therefore central to the threshold structure in Proposition 2.

Because pricing authority and revenue sharing also shape the return to genuine green investment, a selling format can raise genuine greenness and greenwashing at the same time. Section 5 shows this directly by endogenizing both decisions. Figure 1 compares the platform's profit-maximizing format with the greenwashing-minimizing format; solid boundaries show the former and dashed boundaries the latter.

Figure 1 shows three regions that correspond closely to Proposition 2. At low commissions (approximately $\lambda < 0.1$ for the plotted parameters), the platform's profit objective and greenwashing control are aligned on RR; this is consistent with $\hat{\lambda} \in (0.102, 0.134)$ across the plotted trust range. At intermediate commissions (roughly $0.1 < \lambda < 0.5$), AR minimizes greenwashing, whereas the platform may prefer RR, RA, or AA depending on consumer trust, so the two objectives diverge. At higher commissions they realign on AA and then

AR. The final switch to AR at high commissions is a participation effect rather than a new greenwashing incentive: under agency the green supplier retains only $1 - \lambda$ of revenue and can no longer cover the greenness cost, so AA becomes infeasible and the green product must return to reselling.

4.2. Blockchain adoption scenario

We now examine equilibrium outcomes under blockchain verification. The next lemma shows how the per-unit verification cost affects each member's profit across the four selling formats.

Lemma 3. *A higher blockchain cost c_g affects profits as follows:*

- (1) *The platform's profit decreases in RR, RA, and AA; in AR it can increase.*
- (2) *The green supplier's profit decreases in RR, RA, and AR and is unaffected in AA.*
- (3) *The ordinary supplier's profit increases in RR, RA, and AR and is unaffected in AA.*

The platform-profit result in Lemma 3(1) has a distinct mechanism in AR. Because the ordinary product is sold by agency, a higher verification cost can shift demand toward it and increase the platform's commission revenue enough to raise its overall profit; Appendix B gives the exact condition. Specifically, a higher verification cost raises both retail prices but raises the green resale price more than the ordinary agency price, shifting demand toward the ordinary product and expanding the platform's commission base; the resulting increase in ordinary-product commissions can offset both the lost green resale margin and the verification cost. Verified greenness does not automatically translate into a higher green retail price: the green product commands a premium only if $g > \hat{g}^{Bj}$, a format-specific threshold derived in Appendix B. This threshold reflects the interaction of genuine greenness with who controls each retail price and the cross-price response between the two products; green demand must rise enough to overcome the price gap created by the selling formats.

4.3. Comparison and optimal strategy

4.3.1. Joint choice of selling format and blockchain adoption

For a fixed configuration j , the platform's adoption incentive is

$$\Delta_p^j = \pi_p^{Bj} - \pi_p^{Nj}.$$

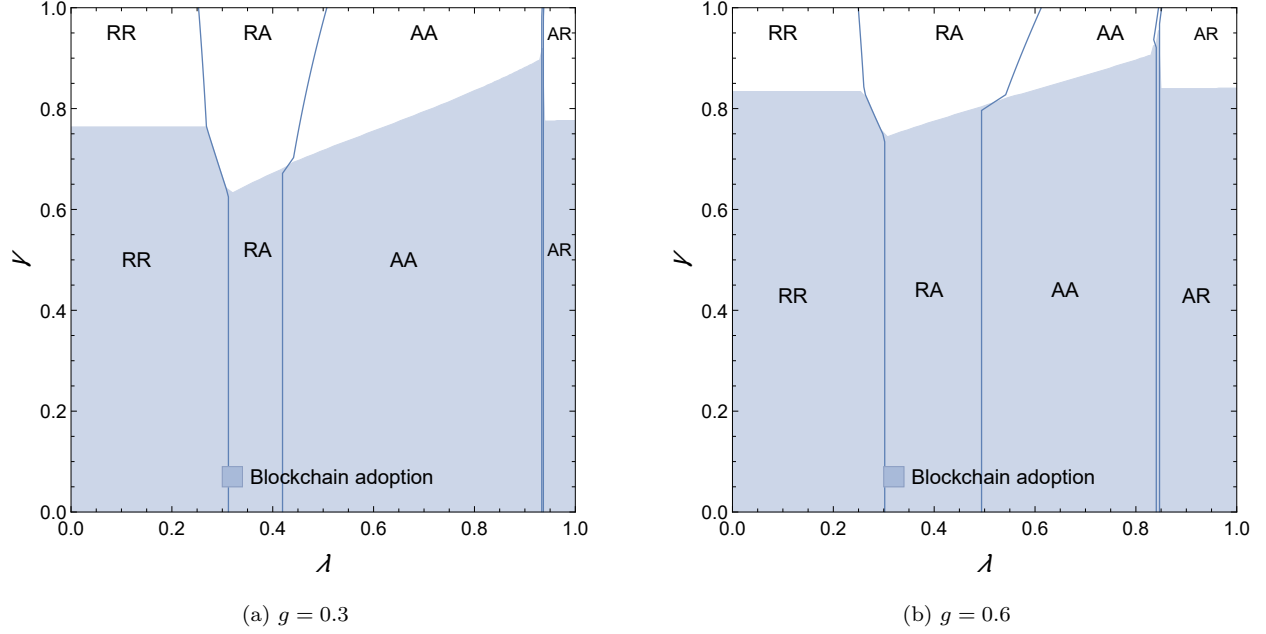


Figure 2: Platform's optimal choice ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 0.4$, $c_g = 0.02$).

Blockchain restores trust, removes demand generated by exaggerated claims, and adds a verification cost. We characterize the adoption regions numerically here; Section 5 derives a common analytical trust threshold once greenness is endogenous.

Figure 2 shows the platform's highest-profit choice of selling format and blockchain adoption at each commission rate and level of consumer trust. Shaded regions indicate blockchain adoption, and solid lines separate optimal selling formats.

Figure 2 shows two threshold patterns. First, blockchain is adopted at low trust and forgone at high trust. For the plotted parameters, the adoption boundary lies roughly between $\gamma = 0.71$ and $\gamma = 0.90$; its position varies with the commission rate and g . When trust is low, unverified claims are heavily discounted, so verified greenness generates enough additional demand to justify the verification cost; when trust is high, unverified claims already create substantial demand and verification mainly removes sales generated by exaggerated claims while adding cost. Second, the selling-format sequence is driven mainly by the commission rate: $RR \rightarrow RA \rightarrow AA$. At very high commissions, however, the green supplier's agency participation constraint makes AA infeasible. This creates the narrow RA band visible near the right edge of Figure 2: over this small interval the platform still keeps the ordinary product under reselling while the green product remains in agency, before the green supplier's

participation constraint tightens further and the platform switches to AR, placing the green product back under reselling.

4.3.2. Consumer surplus and welfare

Following [Singh and Vives \(1984\)](#), [Liu et al. \(2014\)](#), [Wu et al. \(2015\)](#), let $D = (D_o, D_g)^\top$ and $M = \begin{pmatrix} 1 & -b \\ -b & 1 \end{pmatrix}$. Inverse demand has slope $-M^{-1}$, so integrating it up to the quantities consumers choose and subtracting their expenditure gives surplus evaluated at the attributes they perceive:

$$CS = \frac{1}{2} D^\top M^{-1} D = \frac{D_o^2 + 2bD_oD_g + D_g^2}{2(1 - b^2)}. \quad (24)$$

Without verification, consumers base their purchases on $\gamma(1 + \beta)g$ rather than true greenness g . Let $\delta = \gamma(1 + \beta)g - g$ denote this perceived-minus-true gap. Holding the chosen quantities and expenditure fixed, evaluating utility at true greenness shifts the green demand intercept by $-\delta$ and changes surplus by $-\delta(M^{-1}D)_g = -\delta(D_g + bD_o)/(1 - b^2)$. Realized consumer surplus is therefore

$$CS^R = CS - \frac{\delta(D_g + bD_o)}{1 - b^2}. \quad (25)$$

Under blockchain, $\delta = 0$ and the two surplus measures coincide. For a fixed selling format j , write the adoption effect as $\Delta CS^{R,j} = CS^{Bj} - CS^{R,Nj}$. [Figure 3](#) compares perceived CS across configurations and the effect of blockchain adoption.

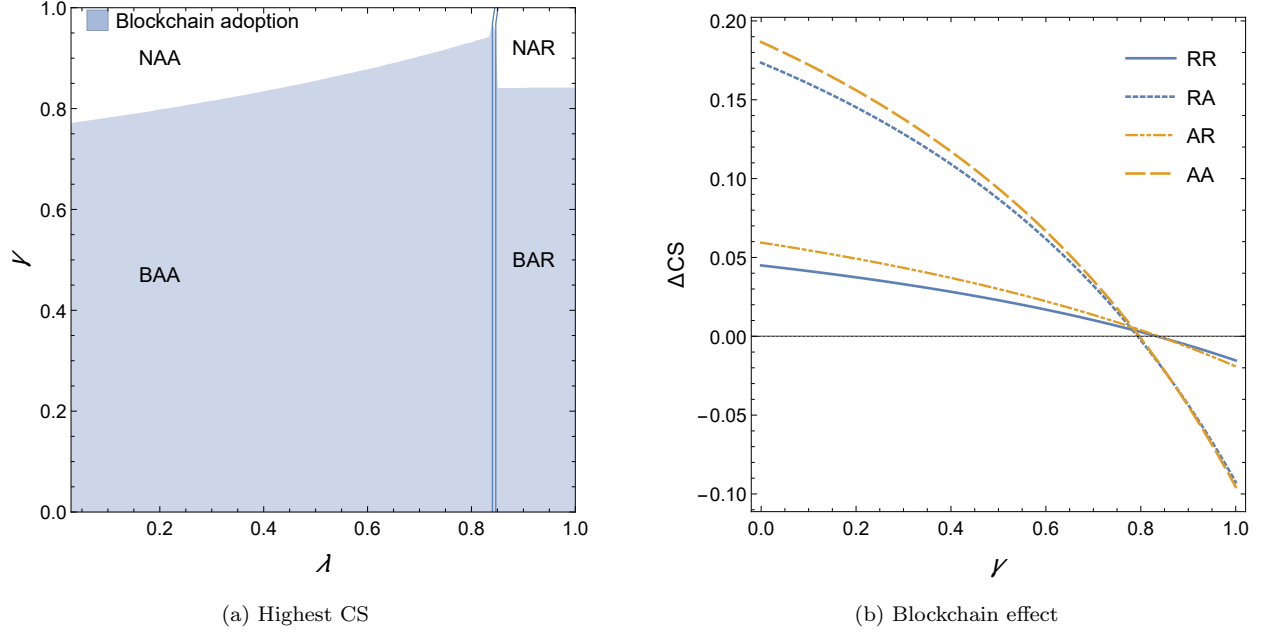


Figure 3: Perceived consumer surplus ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 0.4$, $c_g = 0.02$, $g = 0.6$; panel (b): $\lambda = 0.2$).

Blockchain raises perceived CS when consumer trust is below about 0.8 and lowers it when trust is higher, broadly mirroring the platform's adoption pattern. The effect is larger in RA and AA because the green supplier sets the retail price and can translate verified greenness directly into demand. Over most of the plotted region perceived CS is highest under BAA; at high trust, NAA can instead dominate because verification removes demand that consumers were willing to generate from the unverified claim.

Proposition 3. *Let $c_g = 0$ and suppose both equilibria under configuration j are feasible.*

- (1) *If $\gamma(1 + \beta^{Nj}) > 1$, then $\Delta CS^{R,j} > 0$; if $\gamma(1 + \beta^{Nj}) = 1$, then $\Delta CS^{R,j} = 0$.*
- (2) *If $\gamma(1 + \beta^{Nj}) < 1$, then $\Delta CS^{R,j} < 0$ if and only if $g - \gamma(1 + \beta^{Nj})g < \bar{\delta}^j$, where $\bar{\delta}^j > 0$ is given in Appendix B.*

The sign of δ distinguishes overvaluation from undervaluation of true greenness. Verification raises realized consumer surplus when $\delta > 0$; when $\delta < 0$, it can lower surplus if the understatement is smaller than $\bar{\delta}^j$, while a larger correction can expand demand enough to reverse the effect.

To distinguish changes in efficiency from transfers between consumers and firms, define

$W = CS^R + \pi_o + \pi_g + \pi_p$. Payments between members cancel, so W equals consumers' utility at true greenness minus greenness, greenwashing, and verification costs.

Proposition 4. *Let $c_g = 0$ and suppose both equilibria under configuration j are feasible. Let δ^j be the gap δ at the no-blockchain equilibrium under j , and let $d^j = \partial D^{Bj} / \partial g$, whose components are positive. Then*

$$\Delta W^j = W^{Bj} - W^{Nj} = \theta(\beta^{Nj}g)^2 - \delta^j (d^j)^\top p^{Bj} + \frac{1}{2}(\delta^j)^2 (d^j)^\top M^{-1} d^j,$$

where $p^{Bj} = (p_o^{Bj}, p_g^{Bj})^\top$. If $\gamma(1 + \beta^{Nj}) \leq 1$, then $\Delta W^j > 0$.

Proposition 4 separates the saved greenwashing cost from the effects of correcting beliefs and changing output. When claims understate true greenness, verification raises total welfare. When claims overstate it, verification can protect consumers yet lower welfare if removing claim-driven demand further reduces output under market power. Figure 4 compares the consumer-surplus and welfare effects.

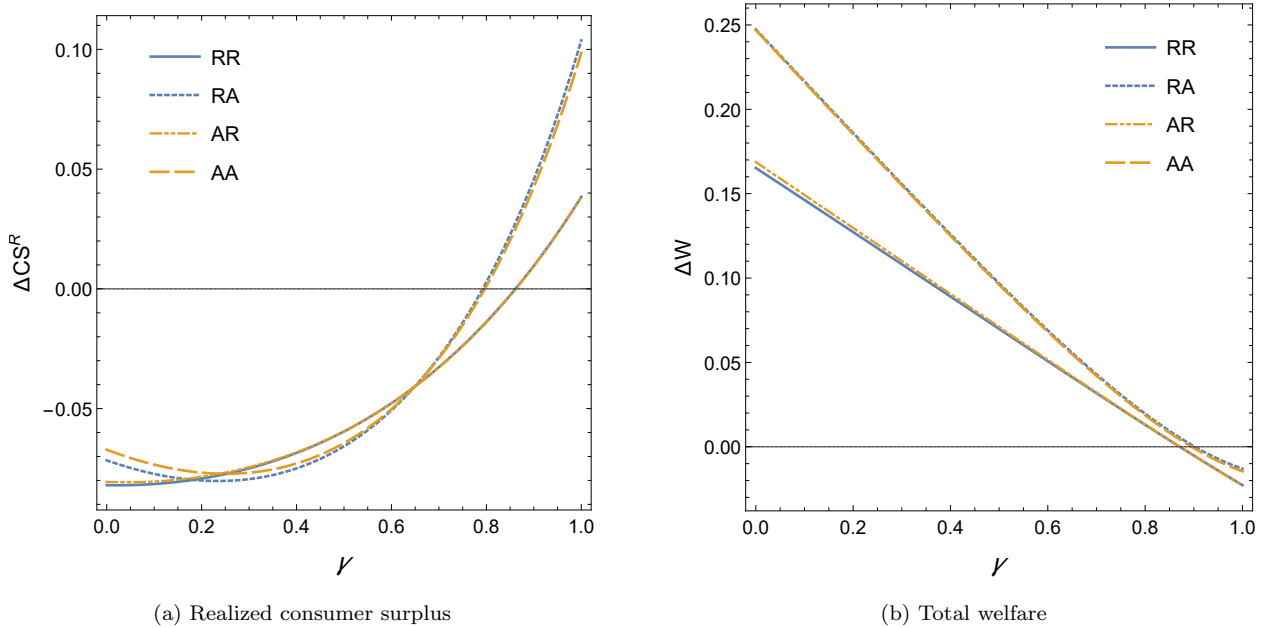


Figure 4: Effect of blockchain on realized consumer surplus and total welfare ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 0.4$, $c_g = 0.02$, $g = 0.6$, $\lambda = 0.2$).

In Figure 4a, blockchain lowers realized CS at low trust because consumers otherwise obtain genuine greenness at discounted prices; at high trust, it raises CS by removing payments

for greenness that is not delivered. The switch is close to $\delta = 0$, with a small displacement due to the positive verification cost used in the figure.

Figure 4b shows that, at $\lambda = 0.2$, blockchain raises total welfare until roughly $\gamma = 0.871$ – 0.902 across formats, while perceived greenness reaches true greenness earlier, around $\gamma = 0.793$ – 0.854 . In RR, which the platform selects at this commission rate, its adoption boundary in Figure 2b lies at $\gamma \approx 0.833$, the perceived and true greenness levels coincide at $\gamma \approx 0.854$, and verification stops raising total welfare at $\gamma \approx 0.871$. Thus, the platform can forgo verification while it still improves total welfare; the consumer-surplus threshold need not coincide with either boundary.

5. Joint Decisions on Genuine Greenness and Exaggeration

In the basic model, product greenness g is exogenous and the green supplier chooses only the exaggeration level β . We now endogenize both environmental decisions. Without blockchain, the green supplier jointly chooses genuine greenness g and the exaggeration level β at Stage 1 together with his price term. With blockchain verification, exaggeration is eliminated ($\beta = 0$), so the supplier chooses only genuine greenness. The timing, contract structure, and absolute-exaggeration cost $\theta(\beta g)^2$ are the same as in the basic model.

As in the basic model, $\beta \geq 0$ is the proportion by which the supplier inflates true greenness: the claim is $(1 + \beta)g$ and perceived greenness is $\gamma(1 + \beta)g$. The cost $\theta(\beta g)^2$ continues to depend on the absolute exaggeration βg ; now both g and β are choices. Without blockchain, the green product's demand and the green supplier's profit become

$$D_g^N = (1 - \rho)a - p_g + bp_o + \gamma(1 + \beta)g, \quad \pi_g^{Nj} = R_g^j - \frac{k}{2}g^2 - \theta\beta^2g^2, \quad (26)$$

where R_g^j denotes the green supplier's revenue under configuration j ($w_g D_g^N$ under reselling and $(1 - \lambda)p_g D_g^N$ under agency).

Lemma 4. *When blockchain is adopted, the equilibrium greenness g^{Bj} is positive if $k > \tilde{k}^{Bj}$ and $c_g < \tilde{c}_g^{Bj}$, where $\tilde{k}^{BRR} = \frac{1}{4-b^2}$, $\tilde{k}^{BRA} = \frac{4(1-\lambda)}{8-b^2(3\lambda+5)}$, $\tilde{k}^{BAR} = \frac{4-b^2(3\lambda+1)}{2(8-b^2(3\lambda+5))}$, and $\tilde{k}^{BAA} = \frac{2(1-\lambda)}{4-b^2}$. The cost thresholds $\tilde{c}_g^{Bj}$ are given in Appendix B; in BAA, no cost condition is needed.*

The condition $k > \tilde{k}^{Bj}$ requires genuine greenness to be sufficiently costly relative to its demand effect. The basic-model value $k = 0.4$ violates this condition in the agency formats at low commission rates (e.g., $\tilde{k}^{BAA} \approx 0.5$ as $\lambda \rightarrow 0$). Without blockchain, the analogous condition applies to the effective greenness-cost coefficient described in Proposition 6. The numerical analysis in this section therefore uses $k = 1.2$, while retaining $\theta = 1.4$ and all other parameter values.

Proposition 5. *Under blockchain, suppose $\lambda \leq \frac{1}{2}$, $k > \tilde{k}^{BRA}$, and all four configurations have interior equilibria. Then $g^{BRA} > \max\{g^{BRR}, g^{BAR}, g^{BAA}\}$ and $g^{BAA} \geq g^{BRR}$, so the lowest greenness is $\min\{g^{BRR}, g^{BAR}\}$.*

RA gives the green supplier direct pricing control while leaving most of his revenue with him when $\lambda \leq 1/2$; the ordinary product's resale price also softens the green supplier's competitive response. This makes genuine investment attractive after verification. Proposition 6(4) shows that, without verification, RA also increases exaggeration. RA can therefore raise both genuine greenness and greenwashing. Wu and Zhao (2027) find that reselling can outperform agency for greenness at higher commissions under green R&D cost sharing.

Without blockchain, the green supplier chooses genuine greenness and exaggeration jointly. The following proposition characterizes this choice analytically.

Proposition 6. *Suppose the no-blockchain game with endogenous greenness has an interior equilibrium under configuration j .*

- (1) $\beta^{Nj} = k/(2\theta)$ for every j .
- (2) *Equilibrium prices, demands, and profits coincide with those of the blockchain game with $c_g = 0$ when its greenness-cost parameter is set to $\frac{2k\theta}{(k+2\theta)\gamma^2}$. In this equivalent blockchain game, equilibrium greenness equals $\gamma(1 + \beta^{Nj})g^{Nj}$.*
- (3) *If $c_g = 0$ and both equilibria are feasible, every member strictly prefers blockchain adoption if and only if $\gamma < \hat{\gamma} \equiv \sqrt{2\theta/(k+2\theta)}$.*
- (4) *If $\lambda \leq \frac{1}{2}$ and Proposition 5 applies to the blockchain game with $c_g = 0$ and greenness-cost parameter $\frac{2k\theta}{(k+2\theta)\gamma^2}$, then RA yields the highest g^{Nj} and the highest absolute exaggeration $\beta^{Nj}g^{Nj}$.*

Proposition 6 characterizes the green supplier's two environmental decisions. Genuine

greenness g improves the product itself, while exaggeration β raises reported greenness without changing the product. Because both raise perceived greenness in the unverified market, the first-order conditions give $\beta = k/(2\theta)$ in every selling format. Relative costs therefore determine proportional exaggeration. Selling format and consumer trust determine equilibrium genuine greenness and, through it, the absolute amount exaggerated βg . A contract that raises genuine greenness can therefore also increase greenwashing. Higher genuine greenness and lower greenwashing are not the same outcome.

Part (2) gives a useful equivalence. Exaggeration lowers the cost of creating perceived greenness, while consumer distrust reduces its market value. In the model, these two effects can be summarized by an adjusted greenness cost. With costless verification, this leads to the same adoption threshold for the platform and both suppliers in every selling format. A positive verification cost lowers the platform's threshold, as Figure 5 shows. Selling contracts determine how much genuine greenness is chosen and how much is exaggerated; verification removes exaggeration from the supplier's choice.

The same logic extends when consumers put different weights on genuine greenness and exaggeration. If genuine greenness is partly verifiable, consumers can place more weight on genuine performance than on an exaggerated claim. The first-order conditions still give a proportional exaggeration that is independent of selling format, and the unverified game can still be written as a verified game with an adjusted greenness cost. The result changes when greenwashing costs depend on sales, for example when fines rise with the quantity sold under a false claim, or when the platform bears a reputational cost. In those cases, the selling format affects exaggeration through equilibrium sales as well as through genuine greenness.

Figures 5 and 6 illustrate these results. In Figure 5, shading marks blockchain adoption, solid lines separate selling formats, and the dotted line marks the costless-verification threshold $\hat{\gamma}$. The shaded adoption boundary lies below $\hat{\gamma}$ because the figure uses a positive verification cost, $c_g = 0.02$: costless verification would be attractive up to the common analytical threshold, whereas a positive cost requires a larger trust deficit before adoption becomes profitable. The small kinks in the adoption boundary occur where the platform's preferred selling format changes, because the profit effect of verification is evaluated under

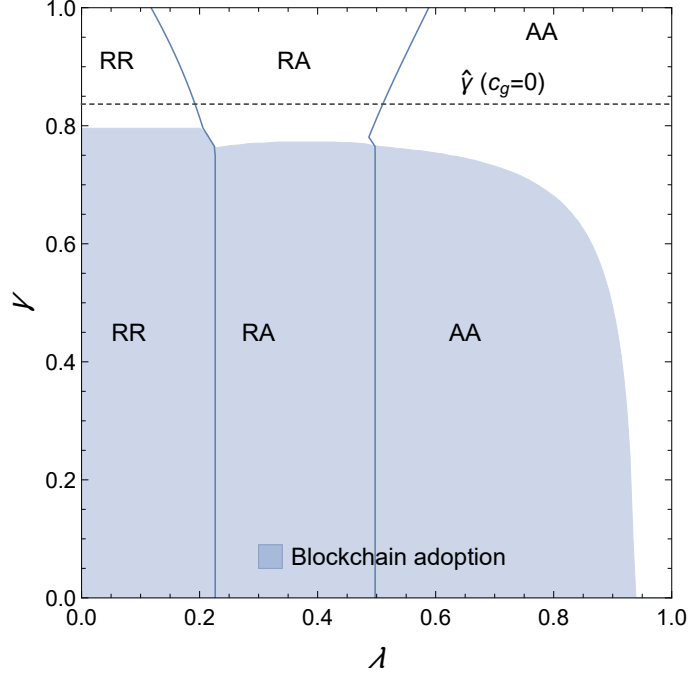


Figure 5: Endogenous greenness: selling format and blockchain adoption ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 1.2$, $c_g = 0.02$).

a different contract on each side of those boundaries.

Figure 6 reports endogenous greenness and absolute exaggeration under the platform's optimal strategy as consumer trust γ varies. The commission rate is common across selling formats. We plot $\lambda = 0.05$, 0.3 , and 0.7 as representative cases with low, intermediate, and high commission rates, in which the platform's preferred format is RR, RA, and AA, respectively. At each trust level, the platform's selling format and blockchain decision are optimized jointly. Blue curves indicate blockchain adoption and orange curves no adoption. Under blockchain (low trust), exaggeration is zero and genuine greenness does not depend on γ . Once trust is high enough that the platform forgoes blockchain, genuine greenness drops discretely, because the supplier substitutes exaggeration for part of his genuine investment. Both βg and g then rise with γ , while the inflation ratio remains $\beta = k/(2\theta) \approx 0.429$, as Proposition 6 predicts. Among the three plotted commission rates, the RA case ($\lambda = 0.3$) yields the highest greenness and absolute exaggeration.

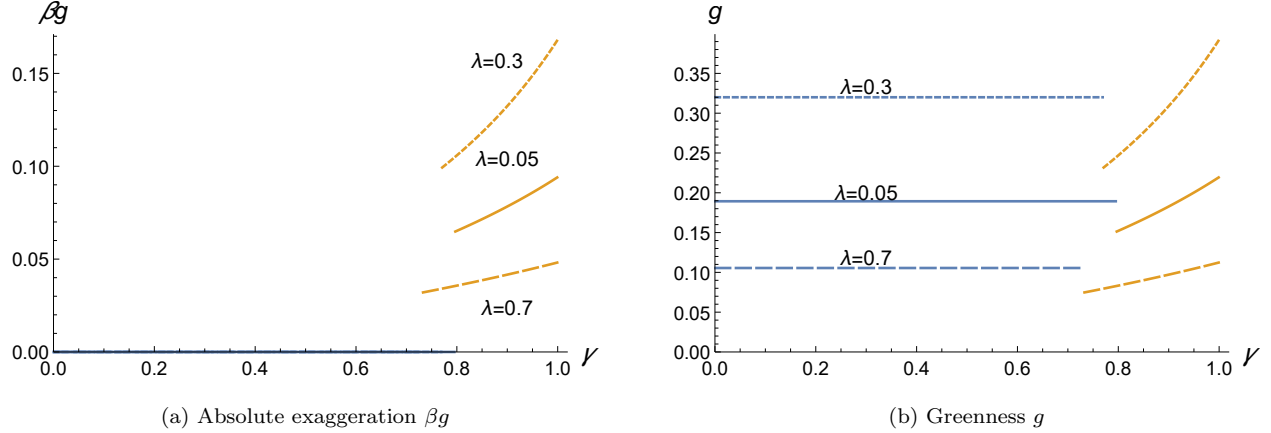


Figure 6: Exaggeration and greenness ($a = 1$, $b = 0.2$, $\rho = 0.3$, $\theta = 1.4$, $k = 1.2$, $c_g = 0.02$; $\lambda = 0.05, 0.3, 0.7$).

6. Conclusions and Managerial Implications

6.1. Summary of findings

This study investigates the joint decisions of an e-commerce platform regarding selling format selection and blockchain adoption in a supply chain with two competing suppliers, one offering an ordinary product and the other a green product. The green supplier may engage in greenwashing when consumers cannot verify true greenness. We analyze four selling format configurations with and without blockchain, and then endogenize both genuine greenness and greenwashing. Our analysis yields three main conclusions.

First, selling formats change greenwashing incentives because pricing authority and revenue sharing change the supplier's profit from exaggeration. When the commission rate is at most one half, RR produces the least greenwashing below a commission threshold and AR produces the least above it; RA produces the most greenwashing. At higher commission rates, AA or AR produces the least greenwashing. The platform's preferred format can therefore differ from the format with the least greenwashing. The same contract can also increase both genuine green investment and exaggeration.

Second, blockchain verification has two effects. It restores trust and removes exaggeration, but it also removes demand generated by exaggerated claims. When claims overstate true greenness, verification raises realized consumer surplus. When consumers moderately undervalue true greenness, verification can lower realized consumer surplus. Total welfare follows a different pattern: verification raises welfare when claims understate true greenness,

but it can lower welfare when exaggerated claims partly offset the low output caused by market power. Consumer protection, platform profit, and total welfare can therefore favor different decisions. In the illustrated parameter settings, the platform adopts blockchain when consumer trust is low and forgoes it when trust is high.

Third, when genuine greenness and exaggeration are chosen together, the optimal proportional exaggeration is determined by the relative costs of genuine investment and exaggeration, not by the selling format. Selling format and consumer trust determine genuine greenness and therefore the total amount exaggerated. Genuine environmental improvement and greenwashing can therefore increase together. Under low commissions, RA produces the highest genuine greenness after verification and the highest absolute exaggeration without verification. With costless verification, the platform and both suppliers change their adoption preference at the same trust threshold.

6.2. Managerial implications

For platform managers. Selling formats change both the return to genuine investment and the profit from exaggeration. At low commissions, RA can raise genuine greenness under truthful reporting and also produce the most greenwashing when claims are unverifiable. Managers should compare platform profit, genuine greenness, and greenwashing separately when choosing a selling format. Verification is attractive when the value of credible green information exceeds its cost and the sales lost when exaggerated claims are removed. When the supplier chooses both genuine greenness and exaggeration, verification can also raise genuine investment by removing exaggeration as an alternative.

For green suppliers. The value of agency selling depends on both pricing control and the share of revenue kept by the supplier, as well as on how the competing product is sold. When verification is costless, the green supplier benefits from verification when consumers place sufficiently low trust in unverified claims. Traceability is therefore most valuable in product categories where consumers discount environmental claims heavily.

For policymakers and regulators. Selling-format incentives matter for greenwashing because the platform’s most profitable format may differ from the format with the least exaggeration. Verification can protect consumers even when it lowers total welfare, and it

can raise total welfare while lowering realized consumer surplus. Policy evaluation should therefore consider consumer protection and total welfare separately. The model also shows that selling-format policies can increase genuine greenness and exaggeration at the same time, so higher green investment alone does not show that greenwashing has fallen.

6.3. Limitations and future research

Several extensions are worth studying. First, the model uses one common exogenous commission rate; supplier-specific or endogenous commissions may change the format rankings. Second, the model has one platform; platform competition may change both selling-format choices and greenwashing incentives. Third, the welfare measure does not include regulation, platform liability, or environmental damage. Fourth, empirical work could test the model once reliable measures of greenwashing and platform transactions are available. Fifth, verification is full and consumer trust is exogenous; imperfect verification and consumer learning could change adoption incentives. Sixth, greenwashing and price terms are chosen at the same stage in the current model; choosing greenwashing earlier could affect rivals' prices. Finally, fixed blockchain costs may lead to different adoption patterns from the per-unit cost studied here.

Appendix A. Optimal decisions and profits

For the four no-blockchain cases below, write $\vartheta = \theta g^2$. These are the fixed- g equilibrium expressions with the absolute-exaggeration cost coefficient substituted exactly.

Case NRR

$$\begin{aligned} w_g^{NRR} &= \frac{2\vartheta(a(2-(2-b)\rho)+2\gamma g)}{2(4-b^2)\vartheta-\gamma^2 g^2}, \quad w_o^{NRR} = \frac{a\rho(\gamma^2 g^2-4(2-b)\vartheta)-4b\vartheta(a+\gamma g)}{2\gamma^2 g^2-4(4-b^2)\vartheta}, \quad \beta^{NRR} = \frac{\gamma g(a(2-(2-b)\rho)+2\gamma g)}{4(4-b^2)\vartheta-2\gamma^2 g^2}, \\ p_g^{NRR} &= \frac{a\rho(b\gamma^2 g^2+4(1-b)(2-b)(2b+3)\vartheta)-12(2-b^2)\vartheta(a+\gamma g)}{4(1-b^2)(\gamma^2 g^2-2(4-b^2)\vartheta)}, \\ p_o^{NRR} &= \frac{a\rho((2b^2-3)\gamma^2 g^2+4(1-b)(2-b)(2b+3)\vartheta)+4b(5-2b^2)\vartheta(a+\gamma g)}{4(1-b^2)(2(4-b^2)\vartheta-\gamma^2 g^2)}, \\ \pi_g^{NRR} &= \frac{\vartheta(8\vartheta-\gamma^2 g^2)(a(2-(2-b)\rho)+2\gamma g)^2-2k(\gamma^2 g^3-2(4-b^2)g\vartheta)^2}{4(\gamma^2 g^2-2(4-b^2)\vartheta)^2}, \quad \pi_o^{NRR} = \frac{(4b\vartheta(a+\gamma g)-a\rho(\gamma^2 g^2-4(2-b)\vartheta))^2}{8(\gamma^2 g^2-2(4-b^2)\vartheta)^2}, \\ \pi_p^{NRR} &= \frac{a^2 \rho^2 (\gamma^4 g^4-8(2-b)(1-b)\gamma^2 g^2 \vartheta+32(1-b)(2-b)^2 \vartheta^2)+16(5b^2+4)\vartheta^2(a+\gamma g)^2-8a\vartheta\rho(a+\gamma g)(3b\gamma^2 g^2+4(1-b)(2-b)^2 \vartheta)}{16(1-b^2)(\gamma^2 g^2-2(4-b^2)\vartheta)^2}. \end{aligned}$$

Case NRA

$$\begin{aligned} w_o^{NRA} &= \frac{a\rho(4\vartheta-b\vartheta(3b\lambda+b-2\lambda+2)-\gamma^2 g^2(1-\lambda))+2b\vartheta(1-\lambda)(a+\gamma g)}{\vartheta(8-b^2(3\lambda+5))-2\gamma^2 g^2(1-\lambda)}, \quad \beta^{NRA} = \frac{\gamma g(1-\lambda)(a(4-(4-3b)\rho)+4\gamma g)}{2\vartheta(8-b^2(3\lambda+5))-4\gamma^2 g^2(1-\lambda)}, \\ p_g^{NRA} &= \frac{\vartheta(a(4-(4-3b)\rho)+4\gamma g)}{\vartheta(8-b^2(3\lambda+5))-2\gamma^2 g^2(1-\lambda)}, \quad p_o^{NRA} = \frac{a\vartheta\rho(12-3b^2(\lambda+1)-2b(\lambda+3))+2b\vartheta(\lambda+3)(a+\gamma g)-3a\gamma^2 g^2(1-\lambda)\rho}{2\vartheta(8-b^2(3\lambda+5))-4\gamma^2 g^2(1-\lambda)}, \end{aligned}$$

$$\begin{aligned}
\pi_g^{NRA} &= \frac{\vartheta(1-\lambda)(a(4-(4-3b)\rho)+4\gamma g)^2(4\vartheta-2b^2\vartheta(\lambda+1)-\gamma^2 g^2(1-\lambda))}{4(2\gamma^2 g^2(1-\lambda)-\vartheta(8-b^2(3\lambda+5)))^2} - \frac{g^2 k}{2}, \\
\pi_o^{NRA} &= \frac{(a\rho(\vartheta(b(3b\lambda+b-2\lambda+2)-4)+\gamma^2 g^2(1-\lambda))-2b\vartheta(1-\lambda)(a+\gamma g))^2}{2(2\gamma^2 g^2(1-\lambda)-\vartheta(8-b^2(3\lambda+5)))^2}, \\
\pi_p^{NRA} &= \frac{1}{4(\vartheta(8-b^2(3\lambda+5))-2\gamma^2 g^2(1-\lambda))^2} [4\gamma^2 g^2 \vartheta^2(b^2(1-\lambda(11\lambda+6))+16\lambda)-4a\gamma g\vartheta(b^3\vartheta(\lambda(18\lambda+13)+1)\rho- \\
&\quad 2b^2\vartheta(1-\lambda(11\lambda+6))(1-\rho)-b\rho(4(7\vartheta\lambda+\vartheta)-\gamma^2 g^2(1-\lambda^2))+32\vartheta\lambda(\rho-1))+a^2(b^4\vartheta^2(1-9\lambda(3\lambda+2))\rho^2+ \\
&\quad 4b^3\vartheta^2(\lambda(18\lambda+13)+1)(\rho-1)\rho+2b^2\vartheta(\gamma^2 g^2(1-\lambda)\rho^2-2\vartheta(11\lambda^2(1-\rho)^2-3\lambda\rho(\rho+4)+6\lambda+\rho(\rho+2)-1))+ \\
&\quad 4b\vartheta(1-\rho)\rho(4(7\vartheta\lambda+\vartheta)-\gamma^2 g^2(1-\lambda^2))+\gamma^4 g^4(1-\lambda)^2\rho^2-8\gamma^2 g^2\vartheta(1-\lambda)\rho^2+16\vartheta^2(4\lambda(1-\rho)^2+\rho^2))].
\end{aligned}$$

Case NAR

$$\begin{aligned}
w_g^{NAR} &= \frac{4\vartheta(4-b^2(3\lambda+1))(a+\gamma g)-4a\vartheta\rho(4-b(3b\lambda+b-2\lambda+2))}{4\vartheta(8-b^2(3\lambda+5))-\gamma^2 g^2(4-b^2(3\lambda+1))}, \\
\beta^{NAR} &= \frac{\gamma g((4-b^2(3\lambda+1))(a+\gamma g)-a\rho(4-b(3b\lambda+b-2\lambda+2)))}{4\vartheta(8-b^2(3\lambda+5))-\gamma^2 g^2(4-b^2(3\lambda+1))}, \\
p_g^{NAR} &= \frac{6\vartheta(4-b^2(\lambda+1))(a+\gamma g)+2a\rho(b(3b+2)\vartheta\lambda-b\gamma^2 g^2\lambda+3(b(b+2)-4)\vartheta)}{4\vartheta(8-b^2(3\lambda+5))-\gamma^2 g^2(4-b^2(3\lambda+1))}, \\
p_o^{NAR} &= \frac{12b\vartheta(a+\gamma g)-2a\rho(6b\vartheta+\gamma^2 g^2-8\vartheta)}{4\vartheta(8-b^2(3\lambda+5))-\gamma^2 g^2(4-b^2(3\lambda+1))}, \\
\pi_g^{NAR} &= \frac{(8\vartheta-\gamma^2 g^2)(w_g^{NAR})^2}{16\vartheta} - \frac{g^2 k}{2}, \\
\pi_o^{NAR} &= \frac{2(1-\lambda)(2-b^2(\lambda+1))(a\rho(6b\vartheta+\gamma^2 g^2-8\vartheta)-6b\vartheta(a+\gamma g))^2}{(\gamma^2 g^2(b^2(3\lambda+1)-4)+4\vartheta(8-b^2(3\lambda+5)))^2}, \\
\pi_p^{NAR} &= \frac{1}{(\gamma^2 g^2(b^2(3\lambda+1)-4)+4\vartheta(8-b^2(3\lambda+5)))^2} [-2a^2(\gamma^4 g^4\lambda\rho^2(b^2(\lambda+1)-2)+2\gamma^2 g^2\vartheta\lambda\rho(-(b^3(9\lambda+7)(1-\rho)- \\
&\quad 2b^2(5\lambda+3)\rho+16b(1-\rho)+16\rho)+2\vartheta^2(b^4(9\lambda(3\lambda+2)-1)(1-\rho)^2-4b^3(\lambda(18\lambda+13)+1)(\rho-1)\rho+4b^2(11\lambda^2\rho^2-3\lambda(3-(6-\rho)\rho)-(4-\rho)\rho+2)-16b(7\lambda+1)(1-\rho)\rho-16(\rho(4\lambda\rho+\rho-2)+ \\
&\quad 1))) + 4a\gamma g\vartheta(b\gamma^2 g^2\lambda\rho(b^2(9\lambda+7)-16)+2\vartheta(-(b^4(9\lambda(3\lambda+2)-1)(1-\rho))-2b^3(\lambda(18\lambda+13)+1)\rho-4b^2(2-9\lambda)(1-\rho)+8b(7\lambda\rho+\rho)+16(1-\rho))) + 4\gamma^2 g^2\vartheta^2(b^4(1-9\lambda(3\lambda+2))-b^2(8-36\lambda)+16)].
\end{aligned}$$

Case NAA

$$\begin{aligned}
\beta^{NAA} &= \frac{\gamma g(1-\lambda)(a(2-(2-b)\rho)+2\gamma g)}{2(4-b^2)\vartheta-2\gamma^2 g^2(1-\lambda)}, p_g^{NAA} = \frac{\vartheta(a(2-(2-b)\rho)+2\gamma g)}{(4-b^2)\vartheta-\gamma^2 g^2(1-\lambda)}, p_o^{NAA} = \frac{a\rho(2(2-b)\vartheta-\gamma^2 g^2(1-\lambda))+2b\vartheta(a+\gamma g)}{2(4-b^2)\vartheta-2\gamma^2 g^2(1-\lambda)}, \\
\pi_g^{NAA} &= \frac{\vartheta(1-\lambda)(4\vartheta-\gamma^2 g^2(1-\lambda))(a(2-(2-b)\rho)+2\gamma g)^2}{4(\gamma^2 g^2(1-\lambda)-(4-b^2)\vartheta)^2} - \frac{g^2 k}{2}, \pi_o^{NAA} = \frac{(1-\lambda)(a\rho(2(2-b)\vartheta-\gamma^2 g^2(1-\lambda))+2b\vartheta(a+\gamma g))^2}{4(\gamma^2 g^2(1-\lambda)-(4-b^2)\vartheta)^2}, \\
\pi_p^{NAA} &= \frac{\lambda}{4(\gamma^2 g^2(1-\lambda)-(4-b^2)\vartheta)^2} [a^2(4\vartheta^2(b^2+2(2-b)^2\rho^2-2(2-b)^2\rho+4)-4\gamma^2 g^2\vartheta(1-\lambda)\rho((2-b)\rho+b)+\gamma^4 g^4(1-\lambda)^2\rho^2)+4a\gamma g\vartheta(2\vartheta(b^2+(2-b)^2(-\rho)+4)+b\gamma^2 g^2(\lambda-1)\rho)+4(b^2+4)\gamma^2 g^2\vartheta^2].
\end{aligned}$$

Case BRR

$$\begin{aligned}
w_g^{BRR} &= \frac{2(a+g)-a(2-b)\rho-(2-b^2)c_g}{4-b^2}, w_o^{BRR} = \frac{b(a+g)+a(2-b)\rho+bc_g}{4-b^2}, \\
p_g^{BRR} &= \frac{3(2-b^2)(a+g)-a(2-b)(1-b)(2b+3)\rho+2(1-b^2)c_g}{2(b^4-5b^2+4)}, p_o^{BRR} = \frac{b(5-2b^2)(a+g)+a(1-b)(2-b)(2b+3)\rho+(b-b^3)c_g}{2(b^4-5b^2+4)}, \\
\pi_g^{BRR} &= \frac{a^2(2-(2-b)\rho)^2-(2-b^2)c_g(-2a(2-b)\rho+4(a+g)-(2-b^2)c_g)+4ag(2-(2-b)\rho)+g^2(4-(4-b^2)^2k)}{2(4-b^2)^2}, \\
\pi_o^{BRR} &= \frac{(b(a+g)+a(2-b)\rho+bc_g)^2}{2(4-b^2)^2}, \\
\pi_p^{BRR} &= \frac{2a^2(1-b)(2-b)^2\rho^2+(1-b^2)c_g(-2(b^2+4)(a+g)+2a(2-b)^2\rho+(4-3b^2)c_g)+(5b^2+4)(a+g)^2-2a(1-b)(2-b)^2\rho(a+g)}{4(4-b^2)^2(1-b^2)}.
\end{aligned}$$

Case BRA

$$\begin{aligned}
w_o^{BRA} &= \frac{2b(1-\lambda)(a+g) + a\rho(4-b(3b\lambda+b-2\lambda+2)) + b(4-b^2(\lambda+3))c_g}{8-b^2(3\lambda+5)}, \quad p_g^{BRA} = \frac{4(a+g) - a(4-3b)\rho - b^2c_g}{8-b^2(3\lambda+5)}, \\
p_o^{BRA} &= \frac{a\rho(-3b^2(\lambda+1) - 2b(\lambda+3) + 12) + 2b(\lambda+3)(a+g) - b(4-b^2(\lambda+1))c_g}{2(8-b^2(3\lambda+5))}, \\
\pi_g^{BRA} &= \frac{1}{2} \left(\frac{(1-\lambda)(2-b^2(\lambda+1))(a(4-3b)\rho - 4(a+g) + b^2c_g)^2}{(8-b^2(3\lambda+5))^2} - g^2k \right), \\
\pi_o^{BRA} &= \frac{(a\rho(b(3b\lambda+b-2\lambda+2) - 4) - 2b(1-\lambda)(a+g) - b(4-b^2(\lambda+3))c_g)^2}{2(8-b^2(3\lambda+5))^2}, \\
\pi_p^{BRA} &= \frac{1}{4(8-b^2(3\lambda+5))^2} [2(2-b^2(\lambda+1))(-a(4-3b)\rho + 4(a+g) + b^2(-c_g))(-a(4-3b)\lambda\rho + 4\lambda(a+g) - \\
&\quad (8-b^2(2\lambda+5))c_g) - (2b(\lambda-1)(a+g) + a\rho(b(3b\lambda+b-2\lambda+2) - 4) - b(4-b^2(\lambda+3))c_g)(2b(3\lambda+ \\
&\quad 1)(a+g) + a\rho(-3(2-b)b\lambda - b(b+2) + 4) - b(12-b^2(3\lambda+7))c_g)].
\end{aligned}$$

Case BAR

$$\begin{aligned}
w_g^{BAR} &= \frac{(a+g)(4-b^2(3\lambda+1)) - a\rho(4-b(3b\lambda+b-2\lambda+2)) - (4-b^2(\lambda+3))c_g}{8-b^2(3\lambda+5)}, \\
p_g^{BAR} &= \frac{3(a+g)(4-b^2(\lambda+1)) - a\rho(12-3b^2(\lambda+1) - 2b(\lambda+3)) + (4-b^2(\lambda+1))c_g}{2(8-b^2(3\lambda+5))}, \quad p_o^{BAR} = \frac{3b(a+g) + a(4-3b)\rho + bc_g}{8-b^2(3\lambda+5)}, \\
\pi_g^{BAR} &= \frac{1}{2} \left(\frac{((a+g)(4-b^2(3\lambda+1)) - a\rho(4-b(3b\lambda+b-2\lambda+2)) - (4-b^2(\lambda+3))c_g)^2}{(8-b^2(3\lambda+5))^2} - g^2k \right), \\
\pi_o^{BAR} &= \frac{(1-\lambda)(2-b^2(\lambda+1))(3b(a+g) + a(4-3b)\rho + bc_g)^2}{2(8-b^2(3\lambda+5))^2}, \\
\pi_p^{BAR} &= \frac{1}{4(8-b^2(3\lambda+5))^2} [2\lambda(2-b^2(\lambda+1))(3b(a+g) + a(4-3b)\rho + bc_g)^2 + ((a+g)(b^2(3\lambda-1) + 4) - \\
&\quad a\rho(-3(2-b)b\lambda - b(b+2) + 4) - (4-3b^2(\lambda+1))c_g)((a+g)(4-b^2(3\lambda+1)) - a\rho(4-b(3b\lambda+b-2\lambda+2)) - (4-b^2(\lambda+3))c_g)].
\end{aligned}$$

Case BAA

$$\begin{aligned}
p_g^{BAA} &= \frac{2(a+g) - a(2-b)\rho}{4-b^2}, \quad p_o^{BAA} = \frac{b(a+g) + a(2-b)\rho}{4-b^2}, \\
\pi_g^{BAA} &= \frac{2a^2(1-\lambda)((b-2)\rho+2)^2 + 8ag(1-\lambda)(2-(2-b)\rho) + g^2(8(1-\lambda) - (4-b^2)^2k)}{2(4-b^2)^2}, \quad \pi_o^{BAA} = \frac{(1-\lambda)(b(a+g) + a(2-b)\rho)^2}{(4-b^2)^2}, \\
\pi_p^{BAA} &= \frac{\lambda(2a^2(2-b)^2\rho^2 + (b^2+4)(a+g)^2 - 2a(2-b)^2\rho(a+g)) - (4-b^2)c_g(2(a+g) - a(2-b)\rho)}{(4-b^2)^2}.
\end{aligned}$$

Appendix B. Proofs

Throughout this appendix, $h = 4 - b^2$, $L = 8 - b^2(3\lambda + 5)$, $T = 4 - b^2(3\lambda + 1)$, and $y = a\rho$. The symbol x denotes the green product's demand intercept excluding greenwashing: $x = a(1 - \rho) + \gamma g$ without blockchain, $x = a(1 - \rho) + g$ under blockchain with exogenous greenness, and $x = a(1 - \rho)$ when greenness is endogenous. These quantities are positive. A prime denotes the derivative with respect to c_g .

Proof of Lemma 1. Backward induction gives the equilibrium expressions in Appendix A with $\vartheta = \theta g^2$. Since $g > 0$, each denominator and the green supplier's second-order condition

has the same sign as its fixed- g counterpart after division by g^2 . In NRR, the denominator of β^{NRR} is positive precisely when $\theta > \gamma^2/[2(4 - b^2)]$; the supplier's Hessian determinant is $g^2(2\theta - \gamma^2/4) > 0$ under this condition. The same substitution gives the other three thresholds in Lemma 1 and their second-order conditions. Positive prices and exaggeration follow on these interior branches.

Proof of Lemma 2. Write each equilibrium absolute exaggeration as $e^{Nj} = \beta^{Nj}g$. Substitution of $\vartheta = \theta g^2$ in Appendix A shows that e^{Nj} is a positive constant times a positive affine function of g ; its denominator is independent of g . Therefore $\partial e^{Nj}/\partial g > 0$. The affine function has a positive intercept from the base green market, so $\beta^{Nj} = e^{Nj}/g$ and $\partial \beta^{Nj}/\partial g < 0$. Derivatives with respect to γ, b, ρ, θ , and λ have the stated signs under the interiority condition, as in the fixed- g proof after replacing the effective cost coefficient by ϑ .

Proof of Proposition 1. We use case NRR as an example. Taking derivatives of w_g^{NRR} , w_o^{NRR} , p_g^{NRR} , p_o^{NRR} , π_g^{NRR} , π_o^{NRR} , and π_p^{NRR} with respect to γ , and evaluating their signs under the condition $\theta > \bar{\theta}^{NRR}$, all derivatives are positive. The other three configurations are verified in the same way from Appendix A.

Proof of Proposition 2. All denominators below are positive in the interior equilibria. In terms of x and y , the four expressions in Appendix A become

$$\begin{aligned}\beta^{NRR} &= \frac{\gamma g(2x + by)}{4h\vartheta - 2\gamma^2 g^2}, & \beta^{NRA} &= \frac{\gamma g(1 - \lambda)(4x + 3by)}{2L\vartheta - 4\gamma^2 g^2(1 - \lambda)}, \\ \beta^{NAR} &= \frac{\gamma g[Tx + 2b(1 - \lambda)y]}{4L\vartheta - \gamma^2 g^2 T}, & \beta^{NAA} &= \frac{\gamma g(1 - \lambda)(2x + by)}{2h\vartheta - 2\gamma^2 g^2(1 - \lambda)}.\end{aligned}$$

Cross multiplication yields $\beta^{NAA} \leq \beta^{NRR}$ if and only if $\lambda \geq 1/2$. The NRA denominator implies $\vartheta > 2\gamma^2 g^2(1 - \lambda)/L$. For $\lambda \leq 1/2$, the cross-product for $\beta^{NRA} - \beta^{NRR}$ is positive because

$$\begin{aligned}2\{2(1 - \lambda)h(4x + 3by) - L(2x + by)\} - byL \\ = 4[8 - 16\lambda + b^2(7\lambda + 1)]x + 3b[(7b^2 - 16)\lambda + b^2 + 8]y > 0.\end{aligned}$$

For $\lambda \geq 1/2$, the cross-product for $\beta^{NRA} - \beta^{NAA}$ is positive because

$$2\{h(4x + 3by) - L(2x + by)\} - byL = 12b^2(1 + \lambda)x + 9b^3(1 + \lambda)y > 0.$$

In both cases the lower bound on ϑ gives the strict inequality. Thus RA cannot minimize.

For AR versus RR, the green product is resold in both modes, and the green supplier's first-order condition for β gives $\beta^{Nj} = \gamma g w_g^{Nj} / (4\vartheta)$ for $j \in \{RR, AR\}$. Let $t = \gamma^2 g^2 / \vartheta = \gamma^2 / \theta$; the positive NRR denominator requires $t < 2h$. Cross multiplication of the expressions above and division by $b\vartheta$ show that $\beta^{NAR} - \beta^{NRR}$ has the sign of

$$[4b(2 + b^2)x + b^2(12 - t)y] - \lambda\{12b(2 - b^2)x + [4(8 - 5b^2) - t(4 - 3b^2)]y\},$$

whose two brackets are the numerator and denominator of $\hat{\lambda}$. The first bracket is positive because $t < 12$. Using $t < 2h$, the second exceeds $12b(2 - b^2)x + 6b^2(2 - b^2)y > 0$. Therefore $\beta^{NAR} < \beta^{NRR}$ if and only if $\lambda > \hat{\lambda}$. Finally, $\hat{\lambda} < 1/2$ if and only if the denominator exceeds twice the numerator, that is,

$$4b(2 - 5b^2)x + [32 - 44b^2 - t(4 - 5b^2)]y > 0.$$

When $b^2 \leq 2/5$, the coefficient of x is nonnegative, and the coefficient of y decreases in t and equals $2b^2(2 - 5b^2) \geq 0$ at $t = 2h$; hence $\hat{\lambda} < 1/2$. Comparing AR with AA for $\lambda \geq 1/2$ completes part (1).

For part (2), the preceding comparisons show that RA exceeds RR and AA when $\lambda \leq 1/2$. To compare RA with AR, use the same t . The sign of $\beta^{NRA} - \beta^{NAR}$ is the sign of

$$2L\{[4 - 8\lambda + b^2(3\lambda + 1)]x + 4b(1 - \lambda)y\} + t b(1 - \lambda)y[8(1 - \lambda) - 3T].$$

This expression is affine in t . It is positive at $t = 0$ for $\lambda \leq 1/2$, and at the upper limit $t = L/[2(1 - \lambda)]$ imposed by the positive NRA denominator it equals $\frac{1}{2}L(4x + 3by)[8(1 - \lambda) - T] > 0$. Hence it is positive throughout the admissible interval. RA therefore has the highest greenwashing level when $\lambda \leq 1/2$.

Proof of Lemma 3. In BRR, $w_g = (2x + by - (2 - b^2)c_g)/h$ and $w_o = (bx + 2y + bc_g)/h$, so

$$\begin{aligned}\frac{\partial \pi_g^{BRR}}{\partial c_g} &= -\frac{(2 - b^2)(2x + by - (2 - b^2)c_g)}{h^2} < 0, \\ \frac{\partial \pi_o^{BRR}}{\partial c_g} &= \frac{b(bx + 2y + bc_g)}{h^2} > 0, \\ \frac{\partial \pi_p^{BRR}}{\partial c_g} &= -\frac{(4 + b^2)x + 4by - (4 - 3b^2)c_g}{2h^2} < 0.\end{aligned}$$

The last inequality is evaluated on the feasible interval: $D_g^{BRR} > 0$ implies $2x + by - (2 - b^2)c_g > 0$, which makes its numerator positive. In BAA, retail prices do not depend on c_g ; hence $\partial\pi_g^{BAA}/\partial c_g = \partial\pi_o^{BAA}/\partial c_g = 0$ and $\partial\pi_p^{BAA}/\partial c_g = -(2x + by)/h < 0$.

In BRA, $p_g = (4x + 3by - b^2c_g)/L$ and $w'_o = b[4 - b^2(\lambda + 3)]/L > 0$. Thus $\pi_g^{BRA} = \frac{1}{2}(1 - \lambda)[2 - b^2(1 + \lambda)]p_g^2 - kg^2/2$ falls with c_g , while $\pi_o^{BRA} = w_o^2/2$ rises with c_g . Substituting $p'_g = -b^2/L$, $p'_o = -b[4 - b^2(1 + \lambda)]/(2L)$, and w'_o into $\pi_p^{BRA} = (p_o - w_o)D_o + (\lambda p_g - c_g)D_g$ gives

$$\begin{aligned}\frac{\partial\pi_p^{BRA}}{\partial c_g} &= -\frac{A_{RA} + B_{RA}c_g}{2L^2}, \\ A_{RA} &= [b^4(10\lambda^2 + 38\lambda + 16) - 64b^2(\lambda + 1) + 64]x \\ &\quad + b[b^4(9\lambda^2 + 34\lambda + 17) - b^2(58\lambda + 66) + 64]y, \\ B_{RA} &= b^2[b^4(-\lambda^2 + 2\lambda + 11) - 28b^2 + 16].\end{aligned}$$

The numerator is affine in c_g . It is positive at $c_g = 0$, because both coefficients of A_{RA} are positive for $b, \lambda \in (0, 1)$, and at the upper bound $c_g = (4x + 3by)/b^2$ implied by $p_g > 0$ it equals $2L\{[8 - b^2(6 + \lambda)]x + b[7 - b^2(5 + \lambda)]y\} > 0$. Hence the platform's profit decreases in c_g .

In BAR, $w'_g = -[4 - b^2(\lambda + 3)]/L < 0$ and $p'_o = b/L > 0$. Hence $\pi_g^{BAR} = w_g^2/2 - kg^2/2$ falls and $\pi_o^{BAR} = \frac{1}{2}(1 - \lambda)[2 - b^2(1 + \lambda)]p_o^2$ rises. The platform exception has an exact sign test:

$$\begin{aligned}\frac{\partial\pi_p^{BAR}}{\partial c_g} &= -\frac{A_{AR} + B_{AR}c_g}{2L^2}, \\ A_{AR} &= [9b^4\lambda^2 + 8b^4\lambda + 3b^4 - 20b^2\lambda - 16b^2 + 16]x \\ &\quad + [8b^3\lambda^2 - 2b^3\lambda - 6b^3 - 8b\lambda + 8b]y, \\ B_{AR} &= -b^4\lambda^2 - 10b^4\lambda - 9b^4 + 12b^2\lambda + 24b^2 - 16.\end{aligned}$$

Thus platform profit rises with c_g precisely when $A_{AR} + B_{AR}c_g < 0$ on the feasible domain; equality gives a stationary point. Finally, the price formulas in Appendix A give $(p'_o, p'_g) = (b/(2h), 1/h)$ in BRR, $(-b[4 - b^2(1 + \lambda)]/(2L), -b^2/L)$ in BRA, $(b/L, [4 - b^2(1 + \lambda)]/(2L))$ in BAR, and $(0, 0)$ in BAA. Hence, as c_g rises, w_o rises and w_g falls where they are defined, both retail prices rise in BRR and BAR, both fall in BRA, and neither changes in BAA.

Price premium under blockchain. The price difference is increasing in g . Setting it to zero gives

$$\begin{aligned}\hat{g}^{BRR} &= a(2\rho - 1) - \frac{(b+1)c_g}{2b+3}, \\ \hat{g}^{BRA} &= \frac{a\rho(-3b^2(\lambda+1) - 2b(\lambda+6) + 20) + 2a(b(\lambda+3) - 4) - b(4 - b(b\lambda + b + 2))c_g}{8 - 2b(\lambda+3)}, \\ \hat{g}^{BAR} &= \frac{1}{3} \left(a \left(\frac{\rho(20 - b(3b(\lambda+1) + 2(\lambda+6)))}{4 - b(b\lambda + b + 2)} - 3 \right) - c_g \right), \\ \hat{g}^{BAA} &= a(2\rho - 1).\end{aligned}$$

If $g > \hat{g}^{Bj}$, the green product commands a price premium.

Proof of Proposition 3. Fix configuration j and let G denote perceived greenness. With $c_g = 0$, the two games differ only through perceived greenness: given the equilibrium greenwashing level β^{Nj} , the price conditions of the no-blockchain game coincide with those of the blockchain game evaluated at $G_N = \gamma(1 + \beta^{Nj})g$. Equilibrium demands in both games are therefore $D(G) = D_0 + Gd$, with $D^{Nj} = D(G_N)$ and $D^{Bj} = D(g)$, where $d = (d_o, d_g)^\top$ collects the constant derivatives of D_o and D_g with respect to g listed in the proof of Proposition 6. Let $\delta = G_N - g$ and $e_2 = (0, 1)^\top$, so that the correction term in (25) equals $\delta e_2^\top M^{-1}D$. Because $CS = \frac{1}{2}D^\top M^{-1}D$ is quadratic in G ,

$$\Delta CS^{R,j} = CS(g) - CS(G_N) + \delta e_2^\top M^{-1}D(G_N) = \delta K^j,$$

where

$$K^j = (e_2 - d)^\top M^{-1}D^{Nj} + \frac{1}{2}\delta d^\top M^{-1}d = (e_2 - d)^\top M^{-1}D^{Bj} + \delta(e_2^\top M^{-1}d - \frac{1}{2}d^\top M^{-1}d).$$

Direct computation shows that both components of $(e_2 - d)^\top M^{-1}$ are positive in every configuration: they equal $b(5 - 2b^2)/[2h(1 - b^2)]$ and $3(2 - b^2)/[2h(1 - b^2)]$ in BRR, $b(\lambda + 3)/L$ and $4/L$ in BRA, $3b/L$ and $3[4 - b^2(1 + \lambda)]/(2L)$ in BAR, and b/h and $2/h$ in BAA. Likewise, $e_2^\top M^{-1}d - \frac{1}{2}d^\top M^{-1}d > 0$ for $b, \lambda \in (0, 1)$. Since equilibrium demands are positive, $K^j > 0$ whenever $\delta \geq 0$, which proves part (1). If $\delta < 0$, then $K^j > 0$ if and only if $|\delta| < \bar{\delta}^j \equiv 2(e_2 - d)^\top M^{-1}D^{Nj}/(d^\top M^{-1}d)$, and $\bar{\delta}^j > 0$. This proves part (2).

Proof of Proposition 4. Summing realized consumer surplus and profits, all payments between members cancel. With $c_g = 0$ and a fixed g , total welfare equals $U_t(D) - \theta(\beta g)^2 - \frac{k}{2}g^2$, where

$U_t(D) = D^\top M^{-1} A_t - \frac{1}{2} D^\top M^{-1} D$ is consumers' utility evaluated at true greenness and A_t is the vector of demand intercepts with perceived greenness equal to g . Completing the square gives $U_t(D) = \text{const} - \frac{1}{2} (D - A_t)^\top M^{-1} (D - A_t)$. As in the proof of Proposition 3, $D^{Nj} = D^{Bj} + \delta^j d^j$, and, since $D = A_t - Mp$ under blockchain, $D^{Bj} - A_t = -Mp^{Bj}$. Hence

$$\Delta W^j = \frac{1}{2} (u + \delta^j d^j)^\top M^{-1} (u + \delta^j d^j) - \frac{1}{2} u^\top M^{-1} u + \theta (\beta^{Nj} g)^2, \quad u = -Mp^{Bj},$$

which simplifies to the stated expression because $(d^j)^\top M^{-1} u = -(d^j)^\top p^{Bj}$. The components of d^j are listed in the proof of Proposition 6 and are positive; M^{-1} is positive definite for $b \in (0, 1)$, and equilibrium prices are positive. If $\delta^j \leq 0$, all three terms are nonnegative and $\theta (\beta^{Nj} g)^2 > 0$, which proves the claim.

Proof of Lemma 4. Under blockchain, the green supplier chooses g at Stage 1 together with w_g (reselling) or p_g (agency). Solving the first-order conditions gives, for example, $g^{BRR} = \frac{a(2-(2-b)\rho)-(2-b^2)c_g}{2[(4-b^2)k-1]}$ and $g^{BAA} = \frac{a(1-\lambda)(2-(2-b)\rho)}{(4-b^2)k-2(1-\lambda)}$. The denominator of g^{Bj} vanishes at $k = \tilde{k}^{Bj}$ and its numerator vanishes at $c_g = \tilde{c}_g^{Bj}$, where $\tilde{c}_g^{BRR} = \frac{a(2-(2-b)\rho)}{2-b^2}$, $\tilde{c}_g^{BRA} = \frac{a(4-(4-3b)\rho)}{b^2}$, and $\tilde{c}_g^{BAR} = \frac{a[T-\rho(4-b(3b\lambda+b-2\lambda+2))]}{4-b^2(\lambda+3)}$; the numerator of g^{BAA} is always positive. The green supplier's Stage-1 Hessian is negative definite if and only if $k > \underline{k}^{Bj}$, where $\underline{k}^{BRR} = \underline{k}^{BAR} = \frac{1}{4}$, $\underline{k}^{BRA} = \frac{1-\lambda}{2-b^2(1+\lambda)}$, and $\underline{k}^{BAA} = \frac{1-\lambda}{2}$; each bound lies below $\tilde{k}^{Bj}$. Hence, for $k > \tilde{k}^{Bj}$, the stationary point is a maximum and $g^{Bj} > 0$ if and only if $c_g < \tilde{c}_g^{Bj}$; in BAA the numerator is always positive. For $\underline{k}^{Bj} < k < \tilde{k}^{Bj}$, a positive stationary point would require $c_g > \tilde{c}_g^{Bj}$, and greenness would then increase with the verification cost; we exclude this counterintuitive region.

Proof of Proposition 5. The equilibrium expressions for genuine greenness are

$$\begin{aligned} g^{BRR} &= \frac{2x + by - (2 - b^2)c_g}{2[(4 - b^2)k - 1]}, \\ g^{BRA} &= \frac{(1 - \lambda)(4x + 3by - b^2c_g)}{Lk - 4(1 - \lambda)}, \\ g^{BAR} &= \frac{Tx + 2b(1 - \lambda)y - [4 - b^2(\lambda + 3)]c_g}{2Lk - T}, \\ g^{BAA} &= \frac{(1 - \lambda)(2x + by)}{(4 - b^2)k - 2(1 - \lambda)}. \end{aligned}$$

The stated bound on k exceeds the denominator thresholds of the other three formats when

$\lambda \leq 1/2$; thus all four denominators are positive. Common feasibility gives $g^{BRR} > 0$, hence

$$0 \leq c_g < \frac{2x + by}{2 - b^2}.$$

For each comparison of RA with RR, AR, or AA, cross multiplication gives a numerator affine in k . At $k = \tilde{k}^{BRA}$, the RA denominator vanishes while each competing denominator is positive, so each cross-product is positive. The coefficients of k , after omitting positive common factors, are respectively

$$\begin{aligned} & 2(1 - \lambda)(4 - b^2)(4x + 3by - b^2c_g) - L[2x + by - (2 - b^2)c_g], \\ & 2(1 - \lambda)(4x + 3by - b^2c_g) - Tx - 2b(1 - \lambda)y + [4 - b^2(\lambda + 3)]c_g, \\ & (4 - b^2)(4x + 3by - b^2c_g) - L(2x + by). \end{aligned}$$

Each is affine in c_g and positive at both endpoints of the admissible interval. At $c_g = 0$, the first two have positive x and y coefficients for $\lambda \leq 1/2$, while the third equals $b\{6b(1 + \lambda)x + [4 + b^2(2 + 3\lambda)]y\} > 0$. At $c_g = (2x + by)/(2 - b^2)$, the first is positive because $(2 - b^2)(4x + 3by) - b^2(2x + by) > 0$. The second, multiplied by $2 - b^2$, has coefficients $16(1 - \lambda) + (16\lambda - 12)b^2 - (3\lambda + 1)b^4$ on x and $b[12 - 8\lambda + (5\lambda - 9)b^2]$ on y , both positive for $0 < b < 1$ and $\lambda \leq 1/2$. The third, multiplied by $2 - b^2$, has coefficients $2b^2[2 + 6\lambda - (2 + 3\lambda)b^2]$ on x and $b[8 - (4 - 6\lambda)b^2 - (1 + 3\lambda)b^4]$ on y , also positive. Therefore RA exceeds the other three formats for every admissible k and c_g .

Finally, direct cross multiplication of AA and RR gives a numerator proportional to

$$(2x + by)(4 - b^2)k(1 - 2\lambda) + (2 - b^2)c_g[(4 - b^2)k - 2(1 - \lambda)] \geq 0.$$

Thus $g^{BAA} \geq g^{BRR}$, so AA cannot minimize; the minimum is the smaller of RR and AR.

Proof of Proposition 6. (1) In every configuration, the green supplier's revenue and the other players' responses depend on g and β only through the claim $(1 + \beta)g$. Let R'_g denote the marginal revenue from perceived greenness. The interior first-order conditions are $\gamma R'_g(1 + \beta) = kg + 2\theta\beta^2g$ and $\gamma R'_gg = 2\theta\beta g^2$. Eliminating R'_g gives $kg = 2\theta\beta g$, hence $\beta = k/(2\theta)$. (2) Fix a target claim $S = (1 + \beta)g$ and write $e = \beta g$, so that $g + e = S$. Minimizing $\frac{k}{2}g^2 + \theta e^2$ subject to $g + e = S$ gives $g = \frac{2\theta}{k + 2\theta}S$ and $e = \frac{k}{k + 2\theta}S$, at cost $\frac{k\theta}{k + 2\theta}S^2 = \frac{k\theta}{(k + 2\theta)\gamma^2}(\gamma S)^2$. A blockchain game with zero verification cost and greenness-cost parameter $\frac{2k\theta}{(k + 2\theta)\gamma^2}$ therefore

has the same objective. Because demand and the other players' payoffs depend on the green supplier's choices only through perceived greenness, the equilibrium prices, demands, profits, and perceived consumer surplus coincide. (3) By (2), $\pi_m^{Nj}(\gamma) = \Pi_m^j\left(\frac{2k\theta}{(k+2\theta)\gamma^2}\right)$, where $\Pi_m^j(k)$ denotes member m 's equilibrium profit under blockchain with endogenous greenness and $c_g = 0$. It suffices to show that Π_m^j is strictly decreasing in k between k and $\frac{2k\theta}{(k+2\theta)\gamma^2}$. At $c_g = 0$, the numerator of g^{Bj} in the proof of Lemma 4 is positive, so positive greenness requires $k > \tilde{k}^{Bj}$. Both endpoints, and hence the whole interval, lie in this region, where g^{Bj} is strictly decreasing in k . Given g , the price conditions coincide with those of the exogenous-greenness game, so $\Pi_m^j(k) = \pi_m^{Bj}(g^{Bj}(k))$ for $m \in \{o, p\}$, with π_m^{Bj} from Appendix A at $c_g = 0$. There, g enters only through $a + g$, and the derivatives with respect to g of all equilibrium prices, platform margins, and demands are positive constants. In BRR, for instance, $\partial w_g/\partial g = 2/h$, $\partial w_o/\partial g = b/h$, $\partial(p_g - w_g)/\partial g = (2 + b^2)/[2(1 - b^2)h]$, $\partial(p_o - w_o)/\partial g = 3b/[2(1 - b^2)h]$, $\partial D_g/\partial g = 1/h$, and $\partial D_o/\partial g = b/(2h)$. In BRA, the corresponding derivatives of p_g , w_o , $p_o - w_o$, D_g , and D_o are $4/L$, $2b(1 - \lambda)/L$, $b(1 + 3\lambda)/L$, $2[2 - b^2(1 + \lambda)]/L$, and $b(1 - \lambda)/L$; in BAR, those of w_g , $p_g - w_g$, p_o , D_g , and D_o are T/L , $(4 - b^2 + 3b^2\lambda)/(2L)$, $3b/L$, $T/(2L)$, and $3b[2 - b^2(1 + \lambda)]/(2L)$; in BAA, those of p_g , p_o , D_g , and D_o are $2/h$, b/h , $2/h$, and b/h . Hence Π_o^j and Π_p^j are strictly decreasing in k . For the green supplier, the envelope theorem gives $d\Pi_g^j/dk = -(g^{Bj})^2/2 + (\partial\pi_g^{Bj}/\partial s_o)(ds_o/dk) < 0$, where s_o is the ordinary supplier's price term: a higher rival price raises green demand, and $ds_o/dk < 0$ by the preceding argument. Therefore $\Pi_m^j(k) > \pi_m^{Nj}$ if and only if $k < \frac{2k\theta}{(k+2\theta)\gamma^2}$, equivalently $\gamma < \sqrt{2\theta/(k+2\theta)} = \hat{\gamma}$, with indifference at equality. (4) By (2), the greenness level in the equivalent blockchain game with zero verification cost and greenness-cost parameter $\frac{2k\theta}{(k+2\theta)\gamma^2}$ equals $\gamma(1 + \beta^{Nj})g^{Nj}$. Part (1) implies that g^{Nj} and $\beta^{Nj}g^{Nj}$ are each the same positive, format-independent multiple of that perceived greenness. Proposition 5 therefore carries both rankings over.

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